Technical Manual

Vehicle Washing Management System (Vehicle Washing)

Microservices Architecture, Data Model, Integration Contracts, Security, and Business Logic

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Version Control

Table 1 records the change history of this document. Each modification must add a row with the version, the date, the description of the change, and the person responsible. It must not be confused with the version control of the source code, described in section 15.

Table 1

Version History of the Technical Manual

Version	Date	Description of the change	Responsible
1.0	2026-10-04	Initial creation of the technical manual (in Spanish) from the project documentation.	Development team
2.0	2026-10-05	Cover data completed, system name adjusted to Vehicle Washing, and description of the migration strategy changed.	Development team
3.0	2026-10-07	Translation into English, APA Style (7th ed.) formatting, and verification against the source code. Corrected: payment-service is .NET 8 (ADR-013); Spring Boot 4 instead of 3; SQL Server 2022; real ports; one Liquibase changelog per service (the single master changelog of version 2.0 was wrong); 47 tables; the gateway does not validate tokens; there is no account lockout; manual operator assignment (ADR-014); English execution codes; real request of the booking endpoint; complete endpoint and event catalog; real project structure; audit trail not populated. Two real diagrams were added.	Development team
3.1	2026-10-09	Update against the code of the eleven repositories as of October 9, 2026 (ADR-011 sections 9 to 12 and ADR-015), and creation of the Spanish version (Manual_tecnico_LavadoVehicular_3145556_ES.md). Updated: 49 tables (booking.schedule_history and notification.booking_tracking); loyalty points per service and points ledger with reversal; promotions as percentage coupons; permission matrix of the fixed roles; notification switches of the user; payment-service events, tests, structure, and container; staff notifications; new endpoints, gateway routes, and error codes; real name of the operations repository; technical debt TD-002, TD-014, TD-015, TD-016, and TD-018 resolved.	Development team
4.0	2026-10-16	Review of the eleven repositories with the system running. Updated: 57 tables with the entity-relationship diagram generated from the database, dictionary of the new tables (account lockout, address, consent, operator profile, and inspection), automatic assignment and operator choice, lockout after failed attempts, inspection reports on the server, business data, health and correlation of payment-service, new endpoints and events, real test results (639 unit and 19 end-to-end), technical debt, and resolved discrepancies.	Development team
4.1	2026-10-17	Review against the code of October 17, 2026. Documented the inspection report of the mobile app (section 10.5), the mobile HTTP client with multipart forms and the *EXPO_PUBLIC_WEB_URL* variable (section 12.7), the *expo-image-picker* and *expo-image-manipulator* dependencies (Table 3), and the rule that ignores a late 401 after logging out.	Development team
4.2	2026-10-19	Continuous integration with GitHub Actions in the 10 repositories (Table 6) and closure of TD-006.	Development team
4.3	2026-10-19	Test results of October 19, 2026 (655 automated, 19 end-to-end, and 42 functional cases).	Development team
4.4	2026-10-20	Review against the code of October 20, 2026: the role change of an account was reverted in the code and the manual does not document it; the rule that an administrator account cannot be deactivated (*ADMIN_CANNOT_BE_DISABLED*) and its 403 error codes were added; it is now stated that the web and the app use neither the customer's rescheduling nor the manual sending of notifications.	Development team
4.5	2026-10-21	Review against the code of October 21, 2026: the token of a closed session stops working (security-service and gateway, sections 9.2, 13.2, and 14.1), automatic close of expired bookings (12.3 and Figure 3), operator assignment lock and single count of bookings without an operator (12.4), receipt of at most 2 MB and cash only with its account active (12.5), web and app changes (10.4, 10.5, and 12.7), tests of October 21 (689 automated and 19 end-to-end), and troubleshooting. The gateway routes missing from Table 5 and the changelog numbering (039 and seeds up to 112) were also corrected.	Development team
4.6	2026-10-21	Change of the rule for bookings that end without being closed, reviewed by the team on the night of October 21: *ExpiredBookingsJob* only cancels the scheduled ones; operations-service (*FollowUpJob*, *FollowUpPolicy*, and *FollowUpService*) moves to NO_SHOW only a confirmed booking without an operator, through the internal route */internal/v1/bookings/{id}/no-show*, and leaves the others in the administrator's "To close" list (12.3, 12.4, 12.8, and Figure 3); new endpoints, ServiceNotStarted and ServiceNotFinished events, notification types, and changelogs 040 and 041 (11.5, 12.6, 13.3, and 13.4); single count of bookings without an operator in the mobile app too and other app changes (12.7); tests of the night of October 21 (707 automated and 19 end-to-end, 16.5); *FOLLOW_UP_** variables (16.2); troubleshooting and discrepancies (Tables 26 and 29).	Development team

Version	Date	Description of the change	Responsible
4.7	2026-10-23	Review against *develop* of October 23, 2026 and version *v1.0.0*: MVVM pattern of the web (10.4); layers, ViewModels, and updates of the mobile app (mandatory APK and EAS Update) (10.5); the operator starts from 15 minutes before and with no later limit (operations-service rules table); **To close** looks back 28 days; the server calculates the payment amount; the variables of October 21 already go through the Compose file (ACO-14 closed); the end-to-end tests run only locally; version control with *v1.0.0* and continuous deployment prepared and not tested (15.1 and 15.4); tests of October 23 (825, and 13 in the mobile app) (16.5). Includes the content of Merari Urbano's *fix/A-03*, *fix/A-05*, *fix/A-08*, *fix/A-15*, and *fix/C-01* branches, checked against the code.	Development team
4.8	2026-10-24	Review against *develop* of October 24, 2026 (128 branches merged), which runs on the public server since that day: the gateway checks the session with any scheme, limits */api/v1/auth/**, and sets per-route timeouts (components and 14.1); window to start up to 4 hours after the end (operations-service rules); *booking.status_changed* and *execution.operator_unassigned* events, operations dead-letter queue, and notification pending deliveries, without *booking.created* (Table 22); OWASP A07 applied; CI and continuous deployment merged into *develop*, the October 24 deployment with *:prev*, *rollback.sh*, and the protection against downgrading (15.4); tests of October 24, 1,192 (16.5).	Development team
4.9	2026-10-26	Payment at booking, reviewed against *develop* of October 26, 2026 and deployed to the public server that day: booking-service queries payment-service through an internal path before cancelling (9.4.1 and 13.3) and consumes *payment.reported* and *payment.confirmed* in the *booking.payment-events* queue (13.4); booking and payment rules (12.3 and 12.5); *paymentMode* when creating the booking and codes *INVALID_PAYMENT_MODE* and *CASH_BOOKING_LIMIT* (13.3); migration numbering with 049 and seed 117 (11.5 and 16.4); new variables (16.2).	Development team
4.10	2026-10-26	Versions of October 26, 2026: *v1.1.0* in the 10 repositories (first version of the mobile app) and *v1.1.1* in the web, the mobile app, payment-service, and infra; since infra *v1.1.1*, payment-service is also deployed from *main* (Technologies table, 15.1, and 15.4).	Development team
4.11	2026-10-26	Continuous deployment verified: first automatic run on the night of October 26, payment-service from *main*, APK *versionCode* 6, and over-the-air update published after comparing the native fingerprint; SonarCloud fixes (15.4).	Development team
4.12	2026-10-27	Changes of October 27, 2026: alternatives without the times that overlap another booking of the vehicle, at most 10 services per booking, 400 for an invalid business hours week, *bookedBy* only for staff, single owners query in customer-service, payment persistence tests in the CI, and counts (12.x, errors table, and 16).	Development team
4.13	2026-10-27	Review against *main* of October 27, 2026: versions *v1.1.1* and *v1.1.2* and *main* protected in the 10 repositories (technology table and 15.1); continuous deployment active and the runs of October 27 (15.4); 58 tables and *notification* with 7 (components, data model, and Annex); tests of the *main* CI, 1,337 with payment's 17 integration tests, and 19 of 19 end-to-end (16.5); the booking life cycle (12.x) and the diagnosis of bookings closed by themselves (16.6) no longer cite *ExpiredBookingsJob*.	Development team
4.14	2026-10-27	payment-service release *v1.1.3* approved and pending publication: credential checks at startup, *LavaRapidoPaymentIT* test database with SQL Server 2022, and 275 tests in the CI (16).	Development team
4.15	2026-11-04	payment-service *v1.1.3*: merged into *main* on October 27 and with tag and release on November 4, 2026, in production since November 4, when the machine was turned on (Technology table, 15.1, and 15.4); tests of the *main* CI, 1,348 with payment's 275 and its four test projects (16).	Development team

Note. Dates are written in the ISO 8601 format (year-month-day). Prepared by the development team.

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Note. Numbering starts at 4 because sections 1 to 3 are the cover, the version control, and the table of contents, which have no number in their title. It is kept this way so that the cross-references between documents by section number (for example, "Installation Manual, section 10") remain valid.

4. Introduction

The Vehicle Washing Management System (Vehicle Washing), internally called *LavaRapido*, is a platform that allows booking washing slots, assigning operators, executing and rating services, and verifying payments made by QR code in a single business. From the technical point of view, it is a set of six domain microservices (five written in Java 21 with Spring Boot 4 and one in C# with .NET 8) behind a gateway, with a SQL Server 2022 database that is shared physically and separated logically by schemas, asynchronous messaging through RabbitMQ, an Angular 21 web client, and a React Native mobile client.

This manual documents how the system is built: the architecture and the decisions that support it, the project structure, the data model, the communication between services, the API, security, and the business rules. It is addressed to developers, architects, technical support staff, and instructors who must maintain or evolve the system.

The architecture relies on widely documented concepts: microservices (Newman, 2021), domain-driven design (Evans, 2004), the hexagonal architecture of ports and adapters (Cockburn, 2005), the publish-subscribe messaging pattern (Hohpe & Woolf, 2003), and the REST architectural style (Fielding, 2000).

4.1 Sources and Conventions of This Document

Version 3.0 was verified against the source code of security-service, customer-service, payment-service, lavarapido-infra, the web application, and the mobile application (*develop* branch, October 5, 2026), and against the documentation of the other services (Vehicle Washing Development Team, 2026a, 2026b, 2026c, 2026d, 2026e, 2026f). Version 3.1 reviewed the eleven repositories (*develop* branch, October 9, 2026). Version 4.0 reviewed the eleven repositories again with the whole system running in Docker (*develop* branch, October 16, 2026): the figures for tables, changelogs, and tests come from queries to the database and from running the tests, not from previous documents. Version 4.5 was reviewed against the code of October 21, 2026. Annex B lists the discrepancies found between the code, the documentation, and the previous versions of the manuals, as the guide for writing manuals requires (Servicio Nacional de Aprendizaje [SENA], 2026). Identifiers (schemas, tables, columns, endpoints, events, and classes) are written exactly as in the code.

5. Objective

5.1 General Objective

To document the architecture, technologies, structure, data model, communication between services, API, security, and business rules of the Vehicle Washing Management System, so that another developer can understand, maintain, and continue the project.

5.2 Specific Objectives

- Describe the architectural style (microservices with logical data ownership) and the internal structure of each service.
- Present the real structure of the repositories and of the client applications.
- Present the nine-schema data model, its dictionary, and its integrity rules.
- Document the synchronous (REST) and asynchronous (RabbitMQ) communication with its endpoints and events.
- Explain the security model: authentication with JWT, authorization by role and ownership, validations, and data protection.
- Leave recommendations for maintenance and testing, and record the technical debt and the discrepancies found.

6. Technical Scope

6.1 What This Manual Covers

- The six backend services (security, customer, booking, operations, payment, and notification) and the API Gateway.

- The web application (Angular) and the mobile application (React Native with Expo) in their structure and their relationship with the backend.
- The SQL Server database, its migrations with Liquibase, and its integrity mechanisms.
- Messaging with RabbitMQ, email through SMTP, and push notifications through Expo.
- The local environment with Docker.

6.2 What This Manual Does Not Cover

- The use of the platform by customers, operators, and administrators (User Manual and Administrator Manual).
- The step-by-step installation (Installation Manual).
- Staging and production environments, which are outside the academic scope.
- Payment gateways, electronic invoicing, and inventory management, which are not part of the system.

7. General Description of the System

7.1 Solution Summary

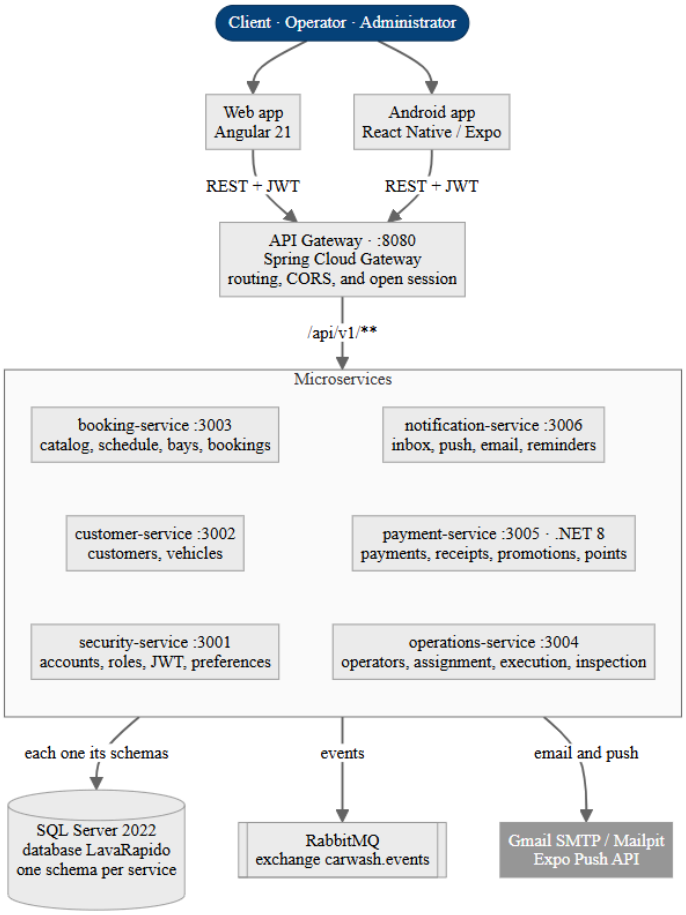
The system serves a single business. Three fixed roles interact with it: the customer, who registers vehicles, books slots, pays, and rates; the operator, who executes the services assigned; and the administrator, who manages bookings, operators, payments, schedules, and reports. Payment does not use an external gateway: the customer transfers from a banking app and uploads the receipt, which the administrator approves or rejects manually (ADR-008). No card data are collected.

7.2 Type of Application and Main Components

It is a distributed client-server system with a microservices backend (Figure 1 and Table 2).

Figure 1

Container Diagram of the System



Note. Prepared by the authors from *docker-compose.yml* and the service catalog. The web application reaches the gateway through *proxy.conf.json* in development.

Table 2

Main Components of the System

Component	Technology	Port	Responsibility
Web application	Angular 21	4200	Interface of the three roles (customer, operator, and administrator).
Mobile application	React Native 0.86 with Expo SDK 57	8081 (Metro)	Interface of the three roles; push notifications and new version notice (APK built with EAS).
api-gateway	Spring Cloud Gateway 2025.1 (webmvc)	8080	Single entry point; routes */api/v1/** by path; CORS; cuts the tokens of closed sessions (since October 21, 2026) with any header scheme (since October 24); limits */api/v1/auth/** to 60 requests per minute per IP and sets 5 s to connect and 60 s to read on each route (since October 24).
security-service	Java 21, Spring Boot 4	3001	People, accounts, roles, sessions, password recovery, profile, and preferences.
customer-service	Java 21, Spring Boot 4	3002	Customer profile, vehicle types, and vehicles.
booking-service	Java 21, Spring Boot 4	3003	Catalog and prices, establishment, schedule, bays, availability, and bookings.
operations-service	Java 21, Spring Boot 4	3004	Operators, shifts, absences, assignment, execution, and ratings.
payment-service	C#, .NET 8, ASP.NET Core, EF Core	3005	Payment accounts with QR, payments with receipt, review, refunds, manual payments, promotions (coupons), and the loyalty points ledger.
notification-service	Java 21, Spring Boot 4	3006	Inbox, push, email, booking reminders, and staff notices.
SQL Server	SQL Server 2022 (container)	1433	Database *LavaRapido*, nine schemas, 58 tables.
RabbitMQ	RabbitMQ 4 with management	5672 and 15672	Domain events (exchange *carwash.events*).
Mailpit or SMTP	Mailpit (development) or Gmail	1025 and 8025	Emails.

Note. Prepared by the authors from the build files and *docker-compose.yml*. The gateway is infrastructure and is not counted as a domain service.

7.3 Architectural Constraints

- The system serves a single business.
- SQL Server is the only database engine; all the services share the database *LavaRapido*, with one owner service per schema.
- No service reads or writes the tables of another; there are no foreign keys between schemas of different services (ADR-003).
- Payment verification is manual; no payment gateway is used (ADR-008).
- The runtime environment is local with Docker.

8. Technologies Used

Table 3

Technologies and Tools of the Project

Component	Technology	Version or detail
Backend language (five services and gateway)	Java (Eclipse Temurin)	JDK 21
Backend framework	Spring Boot	4.0.8: Web MVC, Validation, Security, OAuth2 Resource Server (JWT), Data JPA, Liquibase, Mail, AMQP, Actuator

Component	Technology	Version or detail
API documentation	springdoc-openapi	3.0.3 (Swagger UI only in the *dev* profile)
Gateway	Spring Cloud Gateway (server webmvc)	2025.1
Payment service	.NET, ASP.NET Core, EF Core (SQL Server), JwtBearer, Swashbuckle	.NET 8; EF Core and JwtBearer 8.0.11
Database	Microsoft SQL Server	2022 (image mssql/server:2022-latest); driver mssql-jdbc
Migrations	Liquibase	One changelog per service (SQL formatted); image liquibase 4.29 for payment
Messaging	RabbitMQ	4 with management plugin; topic exchange *carwash.events*
Email	Spring Mail with Mailpit (development) or Gmail	SMTP
Push notifications	Expo Push API	Requires an EAS build on mobile
Web frontend	Angular, Angular Material and CDK, ngx-translate, Chart.js, jsPDF, xlsx, RxJS	21.2.25, 21.2.14, 17, 4.5, 4.2, 0.20, 7.8; TypeScript 5.9
Mobile frontend	React Native, Expo, React, React Navigation, i18next; expo-image-picker and expo-image-manipulator (inspection photos)	0.86.3, SDK 57, 19.2, 7, 26; 57.0; TypeScript 6.0
Containers	Docker and Docker Compose	Docker 24 or later; Compose v2
Testing	JUnit 5 and Spring Boot Test (Java), xUnit (.NET), Vitest (web), Jest with *jest-expo* (mobile, since October 23, 2026)	Through the Maven Wrapper, *dotnet test*, *ng test*, and *npm test*
Version control	Git and GitHub	Branches *main* (tagged versions: *v1.0.0* in 8 repositories on October 23, 2026 and in payment-service on the 24th; *v1.1.0* in the 10 repositories on October 26, *v1.1.1* in the web, the mobile app, payment-service, and infra that same day, and on October 27 *v1.1.1* in customer and booking and *v1.1.2* in the web, the mobile app, payment-service, and infra, and payment-service *v1.1.3*, merged that day and tagged on November 4; *main* protected in all 10), *develop*, *feature/**, and *fix/**
Continuous integration	GitHub Actions	`ci.yml` in the 10 repositories: builds and runs the tests on every push or pull request to *develop* and *main*; `lavarapido-infra` also has the "Desplegar en Azure" workflow, which since October 26, 2026 runs on its own on every approved merge into *main* (continuous deployment active, section 15.4)
Diagrams and contracts	Mermaid (C4 and UML diagrams), OpenAPI 3	Folders 08-uml and 07-api of the documentation

Note. Versions taken from *pom.xml*, *Directory.Build.props*, the *.csproj* files, and *package.json* of each repository.

9. System Architecture

9.1 Architectural Style

The backend adopts a microservices architecture with logical data ownership: each service is independently deployable, is organized around a business capability, and owns its data (Newman, 2021). For academic reasons, the six services share the same SQL Server instance and database, but each schema has exactly one owner service and no service writes to or queries the schema of another (ADR-003). The current decomposition into six services replaced an earlier one with more services (ADR-002, replaced by ADR-009).

Table 4

Services, Owner Schemas, and Tables

Service	Schemas	Tables	Domain depth
security-service	security, audit	12	Medium: authentication, lockout after failed attempts, roles, permissions, password rules, preferences, and consent.
customer-service	customer	3	Medium: ownership of customers and vehicles.
booking-service	catalog, booking	13	High: availability, bay overlap, life cycle, immutable prices, loyalty points per service, and schedule history.
operations-service	execution	11	High: shifts, absences, manual and automatic assignment, execution, rating, operator profile, and inspection reports.
payment-service	payment, promotion	12	High: payment review, refunds, percentage coupons, and points ledger.
notification-service	notification	7	Medium: inbox, idempotent consumption, push, email with pending deliveries and retries (*pending_delivery*), reminders, and booking tracking for staff notices.

Note. Total: 58 tables, counted in the real database on October 27, 2026 (on October 16 there were 57; the one added is *notification.pending_delivery*, from migration 047). Source: changelogs of each service.

9.2 API Gateway

The web and mobile clients call only the gateway (port 8080), so they do not need to know the internal addresses (ADR-005). The gateway routes */api/v1/** by path to the owner service, forwards the method, body, query string, and the headers *Authorization*, *X-Correlation-Id*, and *Accept-Language* without changes, answers CORS (including preflight), and returns 404 for */internal/** and unknown paths. **It does not check the token signature**: each service verifies the JWT itself (ADR-006). Since October 21, 2026 it does confirm that the session of the token is still open: *SessionCheckFilter*, which runs right after CORS, asks security-service with *GET /api/v1/auth/session* and the same *Authorization* header; if the answer is 401, it returns that 401 without passing the request on. It keeps each answer for 30 seconds per SHA-256 fingerprint of the token (never the token itself), forgets the answer of a token when it goes through *POST /api/v1/auth/logout* and, if security-service does not answer within 3 seconds or answers 5xx, lets the request through so that a security-service outage does not bring down the rest of the API. It does not check public routes, the */api/v1/auth/* routes (except logout), or requests without a token. It is configured with *SESSION_CHECK_ENABLED*, *SESSION_CHECK_CACHE_TTL*, and *SESSION_CHECK_TIMEOUT*. It has no database or request rate limit. Table 5 shows the routes.

Table 5

Routes of the API Gateway

Path prefixes	Service
/api/v1/auth/*, */api/v1/users/**, /api/v1/admin/users/*, */api/v1/admin/roles-permissions/**	security-service :3001
/api/v1/vehicles/**, /api/v1/vehicle-types, /api/v1/admin/vehicles	customer-service :3002
/api/v1/catalog/*, */api/v1/schedule/**, /api/v1/establishment, /api/v1/bookings/*, */api/v1/admin/bookings/**, /api/v1/admin/catalog/*, */api/v1/admin/schedule/**, /api/v1/admin/bays/*, */api/v1/admin/establishment, /api/v1/operator/bookings/**	booking-service :3003
/api/v1/admin/operators/*, */api/v1/operators/**, /api/v1/admin/inspections/*, */api/v1/public/inspections/**, /api/v1/admin/assignments/*, */api/v1/operator/services/**, /api/v1/operator/ratings, /api/v1/ratings/**	operations-service :3004
/api/v1/payments/*, */api/v1/payment-accounts, /api/v1/admin/payments/**, /api/v1/admin/payment-accounts/*, */api/v1/admin/payment-methods, /api/v1/admin/promotions/**, /api/v1/loyalty/**	payment-service :3005
/api/v1/notifications/*, */api/v1/admin/notifications/**	notification-service :3006

Note. The addresses are configured with ROUTE_SECURITY_URL, ROUTE_CUSTOMER_URL, ROUTE_BOOKING_URL, ROUTE_OPERATIONS_URL, ROUTE_PAYMENT_URL, and ROUTE_NOTIFICATION_URL; the compose file defines all of them with the container names.

9.3 Internal Architecture of the Services

Java services: hexagonal architecture. The five Java services follow the ports and adapters style (Cockburn, 2005; ADR-007). The domain does not depend on Spring, JPA, SQL Server, or RabbitMQ; the dependencies point toward the domain (Table 6).

Table 6

Layers of the Java Services

Layer	Content	Rule
domain	model (entities and value objects), service (domain policies), event, exception (with a stable code), port/in (use case interfaces), port/out (repositories, publishers, directories).	Does not import Spring or JPA.
application	usecase: services annotated with @Service and @Transactional that implement the input ports.	Coordinates ports; the rules belong to the domain.
infrastructure	adapter/in/web (controllers and DTOs), adapter/out/persistence (JPA entities and repositories), adapter/out/messaging (RabbitMQ), adapter/out/security and notification, config (security, CORS, JWT, OpenAPI, correlation).	Implements the output ports; thin controllers.

Note. Structure verified in security-service and customer-service.

The following excerpt of *AuthenticationService* (security-service) shows a use case that only coordinates ports: it validates the password, applies the rule of the domain (*ensureCanAuthenticate*), records the session, and issues the token.

```
@Service
@Transactional
public class AuthenticationService implements AuthenticateUserUseCase, LogoutUseCase {
    @Override
    public AuthenticationResult login(LoginCommand command) {
        UserAccount account = findByEmail(command.email())
                .filter(found -> passwordHasher.matches(command.password(), found.passwordHash()))
                .orElseThrow(InvalidCredentialsException::new);
        account.ensureCanAuthenticate();          // a disabled account cannot log in
        Instant now = clock.instant();
        account.recordSuccessfulLogin(now);
        accounts.save(account);
        long sessionId = sessions.open(account.id(), now, now.plus(tokenIssuer.timeToLive()),
                command.ipAddress(), command.userAgent());
        IssuedToken token = tokenIssuer.issue(account, sessionId, now);
        return new AuthenticationResult(token.value(), token.expiresAt(), UserAccountView.from(account));
    }
}
```

Note. Abbreviated excerpt of the class; the comment was translated into English. When the email does not exist, the service compares against a dummy hash so that the response time does not reveal which accounts exist.

payment-service: layered (Clean) architecture in .NET. It is organized in four projects with dependencies toward the domain (ADR-013): *PaymentService.Domain* (aggregate *Payment* with its state machine, *PaymentAccount*, *PaymentReceipt*, *PaymentPolicy*, *Promotion*, *LoyaltyTransaction*, and domain events, without NuGet packages), *PaymentService.Application* (use cases by feature: *PaymentCommandService*, *PaymentQueryService*, *BookingPaymentGuard*, and *AmountDueCalculator*; *PaymentAccountService*; *PromotionAdminService* and *CustomerPromotionService*; *LoyaltyService* and *LoyaltyRewardService*; input and output ports in *Ports/In* and *Ports/Out*, including *IBookingDirectory* and *IDomainEventPublisher*), *PaymentService.Infrastructure* (EF Core repositories, *BookingServiceDirectory*, an HTTP client to booking-service, and the RabbitMQ publisher in *Messaging*), and *PaymentService.Api* (controllers, JWT bearer, CORS, *GlobalExceptionHandler*, *EnvFile*, and Swagger). It keeps the same contract as the Java services: HS256 tokens of security-service, RFC 9457 errors with a stable code, */actuator/health*, and configuration from *lavarapido-infra/.env*.

9.4 Communication Between Components

9.4.1 Synchronous Communication With REST

REST is used when the caller needs an answer to continue (ADR-004). The calls between services do not pass through the gateway: the caller forwards the token of the user, so that the target service applies its own access rules (Table 7).

Table 7

Synchronous Calls Between Services

Source	Target and call	Reason
customer-service	security-service GET /api/v1/users/me	Read the personId to create a missing customer profile.
booking-service	customer-service GET /vehicles/{id}, /vehicles, /admin/vehicles	Ownership and type of the vehicle (ADR-012).
operations-service	booking-service /admin/bookings, /operator/bookings, PATCH /operator/bookings/{id}/status, /bookings/me; /internal/v1/bookings, /internal/v1/bookings/{id}, and POST /internal/v1/bookings/{id}/no-show (internal key)	Bookings to assign, with their bay; move them to IN_PROGRESS or COMPLETED; validate the booking being rated; read the booking when BookingConfirmed arrives for the automatic assignment, without a user token (ADR-014); close as NO_SHOW a confirmed booking that ended without an operator (*FollowUpJob*).
operations-service	security-service GET /admin/users?role=OPERATOR; GET /internal/v1/users/{id}/contact (internal key)	Operator accounts and names; customer name of each service and account status (an account that does not exist answers OPERATOR_NOT_FOUND).
payment-service	booking-service GET /bookings/{id}, /bookings/me, /admin/bookings/{id}	Total, status, owner, and loyalty points of the booking; the browser never sends the amount.
notification-service	security-service GET /internal/v1/users/{id}/contact (header X-Internal-Key)	Current email, first name, active status, and notification switches of the user (ADR-011 sections 8 and 12).
notification-service	security-service GET /internal/v1/users/admins (header X-Internal-Key)	Active administrators who receive the staff notices (ADR-011 section 10).
booking-service	payment-service GET /internal/v1/payments/bookings/{bookingId}/reported (header X-Internal-Key, 3 s; *PAYMENT_SERVICE_URL*)	Since October 26, 2026: before cancelling for lack of payment, find out whether the booking already has a payment in review or approved (*PaymentStatusDirectory*, *PaymentServiceDirectory*).
api-gateway	security-service GET /api/v1/auth/session (the same token)	Confirm that the session of the token is still open before passing the request on (since October 21, 2026; section 9.2).

Note. If the target does not answer, the caller responds 503 with a stable code (CUSTOMER_SERVICE_UNAVAILABLE or BOOKING_SERVICE_UNAVAILABLE). Default timeouts: 3 s (5 s in payment-service).

9.4.2 Asynchronous Communication With RabbitMQ

RabbitMQ is used for work that can happen after the original transaction, such as creating the customer profile and the notifications. The events are published after the commit, only when *MESSAGING_ENABLED=true*; otherwise they are written in the log. The catalog of events is in section 13.4.

9.5 Cross-Cutting Concerns

9.5.1 Audit Block and Data Life Cycle

Every domain table carries seven audit columns: *created_at*, *created_by*, *updated_at*, *updated_by*, *deleted_at*, *deleted_by*, and *row_version*. Domain data use logical deletion (*deleted_at*). A trigger `tr_touch_<table>` updates *updated_at* and increments *row_version* in every UPDATE, so the code never writes them.

The design also foresees the table *audit.audit_log*, fed by triggers that read the author from *SESSION_CONTEXT(N'user_id')*. **In the current version, the table exists but nothing writes to it**: the changelogs reviewed do not create audit triggers and the services do not set the session context. It is recorded as a discrepancy (Annex B) and as action ACO-04 of the corrective, preventive, and improvement actions document.

9.5.2 Concurrency and Transactions

The column *row_version* exists, but the entities are not yet mapped with *@Version* (TD-008), so the last write wins in concurrent administrative edits. The critical overlaps are protected by the database: the trigger *tr_booking_bay_overlap* prevents two active bookings from occupying the same bay at overlapping times, and *tr_operator_absence_overlap* prevents overlapping absences. Where two operations at the same time could clash, the Java services take a SQL Server application lock (*sp_getapplock*, owner *Transaction*) that is released when the transaction ends: booking-service uses *booking-schedule* when it creates or reschedules a booking and when it closes the expired ones, and operations-service uses *operations-assignment* in manual and automatic assignment (since October 21, 2026), waiting at most 10 seconds. Each use case opens a local transaction that only writes the schemas of its own service; there are no distributed transactions.

9.5.3 Logging, Correlation, and Errors

The services write their logs to standard output with the level, the service name, and the correlation identifier (*CorrelationIdFilter* in the Java services and *CorrelationIdMiddleware* in payment-service, key *correlationId*, header *X-Correlation-Id*). The header is forwarded in the calls from booking-service to customer-service, from operations-service to booking-service and security-service, and from payment-service to booking-service. It is still missing in the calls from customer-service and notification-service to security-service and in the RabbitMQ messages, and the gateway does not generate it yet (TD-007). Passwords, tokens, recovery codes, and the content of the receipts are never logged.

Errors follow RFC 9457, *Problem Details for HTTP APIs* (Nottingham et al., 2023): the body is *application/problem+json* with a stable *code* that the web application translates with the key `API_ERRORS.<code>`. Stack traces and messages are not included in the responses (*server.error.include-message: never*). Table 8 summarizes the status codes.

Table 8

HTTP Status Codes Used by the Services

Code	Meaning	Examples of code
400	Invalid request or input.	VALIDATION_ERROR, INVALID_PLATE, UNKNOWN_PERMISSION, NOTIFICATION_TYPE_NOT_ALLOWED, RECEIPT_TOO_LARGE
401	Authentication absent, invalid, or expired, or session closed.	UNAUTHORIZED, INVALID_CREDENTIALS
403	Authenticated but not authorized.	FORBIDDEN, ACCOUNT_DISABLED, CANNOT_DISABLE_OWN_ACCOUNT, ADMIN_CANNOT_BE_DISABLED
404	Resource not found or belonging to another user.	VEHICLE_NOT_FOUND, BOOKING_NOT_FOUND
409	Conflict with the current state.	EMAIL_ALREADY_REGISTERED, SLOT_UNAVAILABLE, OPERATOR_BUSY, PAYMENT_ALREADY_REPORTED, TRANSACTION_REFERENCE_REUSED, PROMOTION_ALREADY_REDEEMED
503	Dependency unavailable.	CUSTOMER_SERVICE_UNAVAILABLE, BOOKING_SERVICE_UNAVAILABLE

Note. Codes taken from the exception handlers of the services and the translation file of the web application.

9.5.4 Time and Language

All timestamps are stored in UTC (*DATETIME2(3)*, *SYSUTCDATETIME()*). Booking dates and times are exchanged as local dates of the business (*yyyy-MM-dd* and *HH:mm*) in the zone *BOOKING_ZONE* (America/Bogota), and the responses also carry the UTC instants. The backend returns stable codes, and the clients translate them into Spanish, English, French, or Portuguese.

9.5.5 Patterns Not Adopted

The project decided not to adopt in this version Outbox, Saga, CQRS, Event Sourcing, BFF, service mesh, Redis, database per service, formal circuit breaker, or payment gateway. The most important consequence is the absence of Outbox: an event can be lost if the service stops between the commit and the publication (TD-011).

10. Project Structure

10.1 Repositories

The project is distributed in eleven Git repositories, cloned side by side (Table 9).

Table 9

Repositories of the Project

Repository	Content
lavarapido-infra	docker-compose.yml, .env.example, README, and team guide (GUIA-EQUIPO.md).
lavarapido-security-service, lavarapido-customer-service, lavarapido-booking-service, lavarapido-operation-service (singular; service *operations-service*), lavarapido-notification-service	Java services (Maven Wrapper, Dockerfile, *src/main*, *src/test*).
lavarapido-payment-service	.NET 8 solution, Dockerfile, Liquibase changelogs in *db/*, and its own docker-compose for testing migrations.
lavarapido-api-gateway	Spring Cloud Gateway.
Front-end-proyecto-web	Angular application.
LavadoVehicular-Movil	React Native application with Expo.
vehicle-w-docs	Documentation: governance, requirements, architecture and ADRs, data, API contracts, UML, services, DevOps, quality, UX, operations, and project control.

Note. The repository addresses are listed in the Installation Manual, section 8.

10.2 Structure of a Java Service

The following tree corresponds to security-service; the other Java services follow the same organization (`com.lavarapido.<service>`).

```
lavarapido-security-service/
|-- Dockerfile                     (build with Temurin 21 JDK, run with Temurin 21 JRE)
|-- pom.xml, mvnw, mvnw.cmd
`-- src/
    |-- main/java/com/lavarapido/security/
    |   |-- SecurityServiceApplication.java
    |   |-- domain/
    |   |   |-- model/        UserAccount, Person, EmailAddress, HashedPassword, RoleCode, UserPreference...
    |   |   |-- service/      PasswordPolicy, PasswordRecoveryPolicy
    |   |   |-- event/        UserRegistered, PasswordChanged, EmailChanged, AccountDeactivated
    |   |   |-- exception/    InvalidCredentialsException, WeakPasswordException...
    |   |   `-- port/in, port/out
    |   |-- application/usecase/   AuthenticationService, AccountRegistrationService,
    |   |                          PasswordRecoveryService, UserAdministrationService...
    |   `-- infrastructure/
    |       |-- adapter/in/web/          AuthController, UserProfileController,
    |       |                            UserAdministrationController, InternalUserController, dto/
    |       |-- adapter/out/persistence/ entity/, repository/, *PersistenceAdapter
    |       |-- adapter/out/messaging/   RabbitDomainEventPublisher, LoggingDomainEventPublisher
    |       |-- adapter/out/notification/ EmailResetCodeNotifier, LoggingResetCodeNotifier
    |       |-- adapter/out/security/    BCryptPasswordHasher, JwtTokenIssuer...
    |       `-- config/                  SecurityConfig, JwtConfig, CorrelationIdFilter,
    |                                    InternalApiKeyFilter, DevDataSeeder, OpenApiConfig
    |-- main/resources/
    |   |-- application.yml, application-dev.yml
    |   `-- db/changelog/  db.changelog-master.xml, 001-..., 002-..., 003-..., 020-..., 023-..., 026-..., 030-...,
seed/101-...
    `-- test/java/...      domain, use case, security, and web tests
```

10.3 Structure of payment-service

```
lavarapido-payment-service/
|-- PaymentService.sln, Directory.Build.props (net8.0), Dockerfile
|-- db/changelog/db.changelog-master.yaml, changes/001 ... 020
|-- src/
|   |-- PaymentService.Domain/          Payments/, Promotions/, Loyalty/, Common/
|   |-- PaymentService.Application/     Payments/ (PaymentCommandService, PaymentQueryService,
|   |                                   BookingPaymentGuard, AmountDueCalculator), PaymentAccounts/,
|   |                                   Promotions/, Loyalty/ (LoyaltyService, LoyaltyRewardService),
|   |                                   Ports/In, Ports/Out, Common/
|   |-- PaymentService.Infrastructure/  Persistence/, Booking/BookingServiceDirectory.cs, Messaging/
|   `-- PaymentService.Api/             Controllers/ (Payments, AdminPayments, AdminPaymentAccounts,
|                                       Promotions, Loyalty), Program.cs, Errors/, Configuration/
`-- tests/  PaymentService.Domain.UnitTests, PaymentService.Application.UnitTests,
            PaymentService.Infrastructure.IntegrationTests
```

10.4 Structure of the Web Application

```
Front-end-proyecto-web/src/
|-- app/
|   |-- app.routes.ts            (/, /auth, /client, /operator, /admin, /report/:token)
|   |-- core/
|   |   |-- constants/  api.ts (API_BASE_URL = '/api/v1'), payment-texts.ts (temporary)
|   |   |-- guards/     auth-guard.ts (session and role per area)
|   |   |-- interceptors/ auth-interceptor.ts (Bearer token; session expired -> /auth)
|   |   |-- models/, utils/ (api-error.ts translates the error codes; image-compress.ts keeps the receipt at 2 MB at most)
|   |   `-- services/   auth (session in step across tabs), booking-api, operations-api,
|   |                   payments-api, promotions-api, loyalty-api, role-permissions-api,
|   |                   notifications, vehicles, user-admin, preferences, inspection-api,
|   |                   inspection-photos...
|   |-- features/
|   |   |-- landing/        welcome page
|   |   |-- auth/           login, register, forgot-password (3 steps)
|   |   |-- client/         dashboard, vehicles, reserve, payment, history, notifications, profile, configuration
|   |   |-- operator/       home, schedule, assigned-services, service-history, qualifications, notifications...
|   |   |-- admin/          dashboard, reservations, inspection, payments, management, schedule,
|   |   |                   operators, operator-detail, operator-calendar, reports, settings...
|   |   `-- public-report/  public inspection report
|   |-- layout/, shared/    sidebar, dialogs (modals), components, pipes
|-- assets/i18n/            es.json, en.json, fr.json, pt.json
`-- proxy.conf.json         /api -> http://localhost:8080
```

The application is zoneless and uses signals; each area has a lazy module protected by *authGuard* with the required role (CLIENT, OPERATOR, or ADMIN), and the authentication pages use *guestGuard*.

MVVM pattern. Each page in *features/\<area\>/pages* is the view: a *standalone* component that only displays and forwards events. Its logic is in a ViewModel (*@Injectable*) in *features/\<area\>/viewmodels* or, if several areas share it (profile, settings, create user, sidebar), in *shared/viewmodels*: it keeps the state in signals (*signal*, *computed*), validates the forms, and calls the services in *core/services*, which are the model. The page receives it through *providers* and *inject()*. As of October 23, 2026 there are 36 ViewModels (14 for the administrator, 6 for the operator, 5 for the customer, 3 for authentication, 1 for the landing page, 1 for the public report, and 6 shared); they can be tested with Vitest without mounting the view, and 9 already have their **.viewmodel.spec.ts*. Some administrator ViewModels expose *getters* instead of signals (for example *reservations.viewmodel.ts*). A new rule for a screen goes in its ViewModel, not in the template or the component.

10.5 Structure of the Mobile Application

The mobile application is organized by layers and by role. All the calls go to the gateway through *EXPO_PUBLIC_API_URL*:

```
LavadoVehicular-Movil/src/
|-- app/        config (i18n and es, en, fr, pt locales), providers (session, theme, AppUpdatesCheck,
|               push synchronization), theme
|-- core/       api (fetch client and apiError, which translates the error codes), services (one client
|               per backend service), navigation (RootNavigator: each role only registers its
|               screens), storage (session in secure storage, SecureStore)
|-- modules/    auth, client, operator, admin: each with screens (view), viewmodels (use*.ts hooks
|               with the state and the actions), components, hooks, models, and constants
`-- shared/     components, shared screens (notifications, profile, settings), layouts
                (bottom bar and "More" menu per role), validators, utils, hooks, i18n
```

The app follows the same MVVM idea as the web with React hooks: the screen in *screens* draws and calls its hook in *viewmodels* (for example *useLoginViewModel*, *useInspection*, or *useSettings*; 20 in total), which keeps the state and uses the services in *core/services*. There are 31 screens: 3 for authentication, 8 for the customer, 8 for the operator, and 12 for the administrator. There are still 10 screens or components that call the services without going through a ViewModel (for example *ClientPaymentScreen*, *ClientReserveScreen*, and *OperatorHomeScreen*); it is recorded as technical debt. Push notifications follow the Command pattern (*RegisterDeviceCommand* and *UnregisterDeviceCommand*). Since October 23, 2026 the app has its first tests with *jest-expo* (*src/shared/validators/authValidators.test.ts*, *src/modules/admin/utils/managementUtils.test.ts*, and *src/core/services/__tests__/services.test.ts*, 13 tests), which the CI runs with *npm test -- --ci*.

The administrator's inspection report (RF-036) follows the same organization as the rest of the module: *core/services/operations/InspectionService.ts* calls the same endpoints as the web (*/admin/inspections/...*); *modules/admin/viewmodels/useInspection.ts* keeps the state returned by the server; *modules/admin/services/inspectionPhotos.ts* takes or picks the photos with *expo-image-picker* and shrinks them with *expo-image-manipulator* (1600 px on the longer side, JPEG), because the server accepts at most 2 MB per photo; *modules/admin/components/inspection* holds the card of each phase, the finding form, and the enlarged photo; and *modules/admin/screens/AdminInspectionScreen.tsx* is the screen, opened from the booking detail (navigation route *AdminInspection*). The texts of this module are in *modules/admin/constants/inspectionTexts.ts*, in the four languages.

Since October 21, 2026, *core/services/payments/receiptImage.ts* compresses the payment receipt to at most 2 MB before uploading it, *shared/components/feedback/ImageZoomModal.tsx* enlarges the QR of the payment account, and *app/providers/AppUpdatesCheck.tsx* reads */app/version.json* when the installed APK opens: if there is a higher *versionCode*, it shows a

screen on top of everything that cannot be closed and only lets the user download the new APK; if the server does not answer, the app goes on (*core/services/appUpdates/publishedApk.ts*).

Updates (APK and EAS Update). The app receives changes in two ways, decided by *app/providers/AppUpdatesCheck.tsx*. If the server publishes an APK with a higher *versionCode* (*/app/version.json*), the app is blocked until it is downloaded: it is the only way to deliver native changes (a native package, permissions, icon, or *app.json*); it is built with *eas build -p android --profile preview* and published with *npm run publish:apk*. Changes to screens and logic only arrive over the internet with EAS Update (*expo-updates*, *preview* channel, *runtimeVersion* with the *fingerprint* policy, and a check when the app opens, *checkAutomatically: ON_LOAD*): the app downloads the update and offers to restart; they are published with *npm run publish:update -- "what changed"* (*scripts/publish-update.mjs*, which sets the same backend address as the APK). An update over the internet does not reach an APK with another *runtimeVersion*. The app version is 1.1.0 (*versionCode* 5).

11. Database

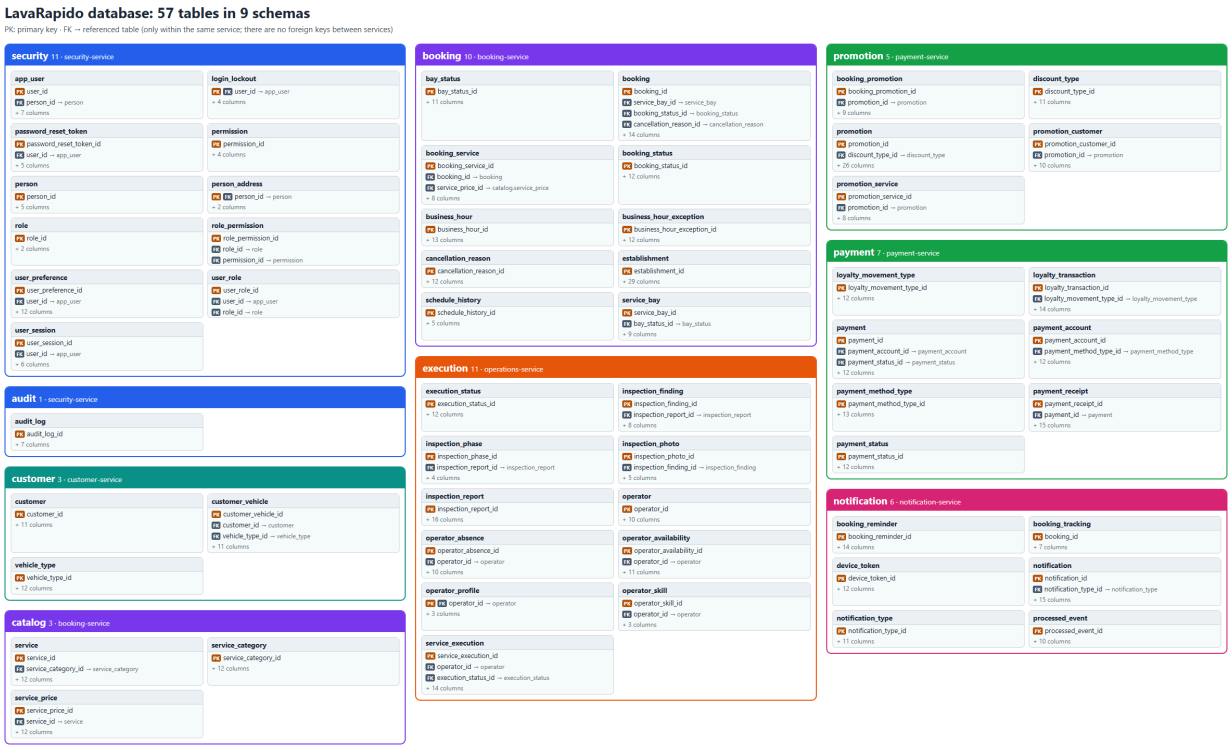
The system uses a single SQL Server 2022 database called *LavaRapido*, divided into nine schemas owned by six services, with 58 tables.

11.1 Entity-Relationship Diagram

Figure 2 was generated directly from the *LavaRapido* database (catalog views *sys.tables*, *sys.columns*, and *sys.foreign_keys*) with all the services started. It shows each table inside its schema, its primary keys, and its foreign keys with the table they point to; the other columns are summarized as a number. References between services (for example, *execution.operator.user_id* to *security.app_user*) do not appear because they have no foreign key (Table 10).

Figure 2

Entity-Relationship Diagram of the System



11.2 Main Tables

11.2.1 Schema Ownership and Foreign Key Rules

Foreign keys are allowed only inside the boundary of a service. Between services, the references are indexed columns whose value the application validates before writing (ADR-003). For example, *booking.booking_service.service_price_id* is a foreign key to *catalog.service_price* because both schemas belong to booking-service, whereas *payment.payment.booking_id* is not (Table 10).

Table 10

References Between Services Without a Foreign Key

Column	Points to	Owner of the target
customer.customer.person_id, customer.customer.user_id	security.person, security.app_user	security-service
catalog.service_price.vehicle_type_id	customer.vehicle_type	customer-service
booking.booking.customer_vehicle_id	customer.customer_vehicle	customer-service
booking.booking.booked_by	security.app_user	security-service
execution.operator.user_id	security.app_user	security-service
execution.service_execution.booking_service_id	booking.booking_service	booking-service
promotion.promotion_service.service_id	catalog.service	booking-service
promotion.promotion_customer.customer_id, promotion.booking_promotion.booking_id	customer.customer, booking.booking	customer-service, booking-service
payment.payment.booking_id, payment.loyalty_transaction.booking_id	booking.booking	booking-service
payment.payment.approved_by, payment.payment_receipt.uploaded_by, reviewed_by	security.app_user	security-service
payment.loyalty_transaction.customer_id	security.app_user (holds the user_id of the customer, ADR-015)	security-service
notification.notification.user_id and the user_id of device_token and booking_reminder	security.app_user	security-service

Note. Prepared by the authors from the changelogs and the data models of the services.

11.2.2 Modeling Conventions

- **Surrogate keys:** IDENTITY(1,1) integers (BIGINT in high-growth tables, INT or SMALLINT in catalogs), not UUIDs.
- **Audit block:** seven columns in every domain table, except the shared security core.
- **Logical deletion:** every query must include *deleted_at IS NULL*; business uniqueness rules use filtered indexes with that condition (for example, the license plate is unique only among active vehicles).
- **Time:** DATETIME2(3) in UTC.
- **Names:** single-word schemas; tables in singular snake_case; primary keys `<table>_id`; constraints *pk_*, *fk_*, *uq_*, *ck_*; indexes *ix_* and *ux_*; triggers *tr_* and *tr_touch_*. *[role]*, *[permission]*, and *[action]* go between brackets because they are reserved words.
- **Catalogs:** separate tables with the shape *code*, *name*, *display_order*, and *is_active*, so that a foreign key prevents mixing statuses of different entities.
- **Status codes:** in English in the database (SCHEDULED, CONFIRMED, IN_PROGRESS, COMPLETED, CANCELLED, NO_SHOW; PENDING, IN_REVIEW, APPROVED, REJECTED, REFUNDED; PENDING, IN_PROGRESS, PAUSED, COMPLETED, WITH_ISSUE). The payment method codes remain in Spanish (EFECTIVO, NEQUI, DAVIPLATA, TRANSFERENCIA).

11.2.3 Tables by Schema

Table 11

Tables of the System by Schema

Schema (owner)	Tables
security (security-service)	person, person_address, app_user, role, permission, role_permission, user_role, user_session, password_reset_token, user_preference, login_lockout
audit (security-service)	audit_log
customer (customer-service)	vehicle_type, customer, customer_vehicle
catalog (booking-service)	service_category, service, service_price
booking (booking-service)	establishment, business_hour, business_hour_exception, bay_status, service_bay, booking_status, cancellation_reason, booking, booking_service, schedule_history
execution (operations-service)	execution_status, operator, operator_availability, operator_absence, service_execution, operator_profile, operator_skill, inspection_report, inspection_phase, inspection_finding, inspection_photo
promotion (payment-service)	discount_type, promotion, promotion_service, promotion_customer, booking_promotion
payment (payment-service)	payment_method_type, payment_account, payment_status, payment, payment_receipt, loyalty_movement_type, loyalty_transaction
notification (notification-service)	notification_type, notification, device_token, processed_event, booking_reminder, booking_tracking

Note. Prepared by the authors from the migration strategy and the data models of the services.

11.3 Data Dictionary

The dictionary lists, for each table, its columns, types, nullability, and constraints. To keep it readable, the standard audit block (seven columns) is summarized as "audit block". The columns of security, customer, and payment were verified in the changelogs; the others come from the data models of the documentation.

Table 12

Data Dictionary of the Security and Audit Schemas

Table	Field	Type	Null	Constraint or description
security.person	person_id	BIGINT	No	PK
	document_number	NVARCHAR(20)	No	Unique (uq_person_document)
	first_name, last_name	NVARCHAR(60)	No	—
	phone	NVARCHAR(20)	Yes	—
	email	NVARCHAR(120)	Yes	—
security.app_user	user_id	BIGINT	No	PK
	person_id	BIGINT	No	FK to person; unique
	username	NVARCHAR(60)	No	Unique; it is the email
	password_hash	NVARCHAR(255)	No	BCrypt hash
	is_active	BIT	No	Default 1
	last_login	DATETIME2(3)	Yes	—
	terms_accepted_at, data_policy_accepted_at	DATETIME2(3)	Yes	Date the Terms and the Data Policy were accepted at registration (changelog 039); null for older accounts
	created_at	DATETIME2(3)	No	Account creation date (changelog 036)
security.role	role_id, code, name	SMALLINT, NVARCHAR(30), NVARCHAR(60)	No	PK; code unique (ADMIN, OPERATOR, CLIENT)

Table	Field	Type	Null	Constraint or description
security.permission	permission_id, code, name, resource, action	SMALLINT, NVARCHAR	No	PK; code unique; catalog loaded by changelog 026
security.role_permission	role_permission_id, role_id, permission_id	BIGINT, SMALLINT, SMALLINT	No	FKs; unique (role_id, permission_id); informative matrix edited by the administrator (ADR-015)
security.user_role	user_role_id, user_id, role_id	BIGINT, BIGINT, SMALLINT	No	FKs; unique (user_id, role_id)
security.user_session	user_session_id, user_id	BIGINT	No	PK; FK to app_user
	refresh_token_hash	NVARCHAR(255)	Yes	Not used (no refresh tokens)
	started_at, expires_at	DATETIME2(3)	No	—
	revoked_at	DATETIME2(3)	Yes	Set at logout
	ip_address, user_agent	NVARCHAR(45), NVARCHAR(300)	Yes	—
security.password_reset_token	password_reset_token_id, user_id	BIGINT	No	PK; FK to app_user
	token_hash	NVARCHAR(255)	No	Unique; BCrypt hash of the code
	requested_at, expires_at	DATETIME2(3)	No	ck: expires_at > requested_at
	used_at, ip_address	DATETIME2(3), NVARCHAR(45)	Yes	—
security.user_preference	user_preference_id, user_id	BIGINT	No	PK; FK and unique user_id
	theme	NVARCHAR(20)	No	green-light, green-dark, pink, or pink-dark
	language	NVARCHAR(5)	No	es, en, fr, or pt
	notifications_enabled	BIT	No	Default 1; push switch
	email_reminders_enabled, promotions_enabled	BIT	No	Default 1 (changelog 030); audit block
security.person_address	person_id, address	BIGINT, NVARCHAR(200)	No	PK and FK to person; contact address of the profile (changelog 035)
security.login_lockout	user_id, failed_count, first_failed_at, locked_until	BIGINT, INT, DATETIME2(3)	No/Yes	PK and FK to app_user; failed attempts and temporary lockout (changelog 034)
audit.audit_log	audit_log_id	BIGINT	No	PK
	entity, entity_id, action	NVARCHAR(60), BIGINT, NVARCHAR(20)	No	—
	previous_data, new_data	NVARCHAR(MAX)	Yes	JSON text
	changed_by	BIGINT	Yes	—
	changed_at	DATETIME2(3)	No	Default SYSUTCDATETIME()

Note. Source: changelogs 001, 002, 003, 020, 026, 030, 034, 035, 036, and 039 of security-service. Changelog 023 created *security.custom_role* and 026 dropped it (ADR-015).

Table 13

Data Dictionary of the Customer Schema

Table	Field	Type	Null	Constraint or description
customer.vehicle_type	vehicle_type_id	SMALLINT	No	PK; ids 1 to 6 fixed
	code, name	NVARCHAR(30), NVARCHAR(60)	No	CAR, SEDAN, SUV, PICKUP, TRUCK, MOTO
	size_factor	DECIMAL(4,2)	No	> 0 (0.80 to 1.35); pricing suggestion
	display_order, is_active	SMALLINT, BIT	No	Audit block
customer.customer	customer_id	BIGINT	No	PK
	person_id	BIGINT	No	Unique; reference to security.person
	user_id	BIGINT	Yes	Unique among active rows; the JWT *sub*
	loyalty_points	INT	No	≥ 0; display copy
	customer_since	DATE	No	Audit block
customer.customer_vehicle	customer_vehicle_id, customer_id	BIGINT	No	PK; FK to customer
	license_plate	NVARCHAR(10)	No	Unique among active vehicles (ux_customer_vehicle_plate_active)
	vehicle_type_id	SMALLINT	No	FK to vehicle_type
	brand, model, color	NVARCHAR(50), NVARCHAR(50), NVARCHAR(30)	Yes	Audit block

Note. Source: changelogs 004, 014, 015, and seed 102 of customer-service.

Table 14

Data Dictionary of the Catalog and Booking Schemas

Table	Field	Type	Null	Constraint or description
catalog.service_category	service_category_id, code, name, description, display_order, is_active	SMALLINT, NVARCHAR	No/Yes	LAVADO, POLICHADO, DETALLADO, INTERIOR, PAQUETES
catalog.service	service_id, code, name, description, service_category_id, is_active	INT, NVARCHAR	No/Yes	FK to service_category; BASIC, PREMIUM, FULL
	loyalty_points	INT	No	Default 0 (changelog 022); points the customer earns when the payment is approved
catalog.service_price	service_price_id, service_id, vehicle_type_id	BIGINT, INT, SMALLINT	No	FK to service; one current price per pair (ux_service_price_current)
	price, estimated_minutes	DECIMAL(12,2), SMALLINT	No	Immutable (tr_service_price_immutable)
	valid_from, valid_to	DATE	No/Yes	Audit block
booking.establishment	establishment_id, legal_name, trade_name, tax_id, address, phone, email, logo_url, singleton; business data (changelog 032): business_type, founded_on, legal_representative, legal_representative_document, website, tax_regime, ciiu_activity, dian_resolution, invoice_prefix, invoice_range, instagram, facebook, support_line, service_hours_note	SMALLINT, NVARCHAR, DATE, BIT	No/Yes	A single row (unique singleton); edited through GET and PUT /api/v1/admin/establishment
booking.business_hour	business_hour_id, day_of_week, opens_at, closes_at, break_starts_at, break_ends_at, is_active	SMALLINT, TIME(0), BIT	No/Yes	One row per day (1 = Monday)

Table	Field	Type	Null	Constraint or description
booking.business_hour_exception	business_hour_exception_id, exception_date, is_closed, opens_at, closes_at, reason	INT, DATE, BIT, TIME(0), NVARCHAR(120)	No/Yes	Unique date
booking.bay_status	bay_status_id, code, name, accepts_bookings, display_order	SMALLINT, NVARCHAR, BIT	No	ACTIVE, MAINTENANCE, INACTIVE
booking.service_bay	service_bay_id, code, name, bay_status_id	SMALLINT, NVARCHAR, SMALLINT	No	FK to bay_status; four bays in the seed
booking.booking_status	booking_status_id, code, name, is_final, display_order, is_active	SMALLINT, NVARCHAR, BIT	No	Ids 1 to 6; 5 = CANCELLED
booking.cancellation_reason	cancellation_reason_id, code, name, is_customer_fault, display_order, is_active	SMALLINT, NVARCHAR, BIT	No	—
booking.booking	booking_id, customer_vehicle_id, service_bay_id, booking_status_id, booked_by	BIGINT, SMALLINT	No/Yes	FKs to service_bay and booking_status; overlap trigger
	scheduled_start, scheduled_end	DATETIME2(3)	No	UTC
	cancellation_reason_id	SMALLINT	Yes	Only with status 5 (ck_booking_cancel)
	points_redeemed, points_discount_amount, notes	INT, DECIMAL(12,2), NVARCHAR(300)	No/Yes	Points stay at 0; no total column
booking.booking_service	booking_service_id, booking_id, service_price_id, quantity	BIGINT, SMALLINT	No	FKs to booking and catalog.service_price
booking.schedule_history	schedule_history_id, entity_type, title, detail, changed_at, changed_by	BIGINT, NVARCHAR, DATETIME2(3)	No/Yes	Log of the changes of schedule, exceptions, and bays (changelog 021)

Note. Source: changelogs 005, 018, 021, 022, and 024 of booking-service. Changelog 024 removed *service_bay.operator_id* (ADR-015). Each table also has the audit block, except the history log.

Table 15

Data Dictionary of the Execution Schema

Table	Field	Type	Null	Constraint or description
execution.execution_status	execution_status_id, code, name, is_final, display_order, is_active	SMALLINT, NVARCHAR, BIT	No	PENDING, IN_PROGRESS, PAUSED, COMPLETED, WITH_ISSUE
execution.operator	operator_id, user_id, hired_on, is_active	INT, BIGINT, DATE, BIT	No	user_id unique; created automatically for OPERATOR accounts
execution.operator_availability	operator_availability_id, operator_id, day_of_week, starts_at, ends_at, is_active	INT, SMALLINT, TIME(0), BIT	No	One row per operator and day; starts_at < ends_at
execution.operator_absence	operator_absence_id, operator_id, starts_at, ends_at, reason	INT, DATETIME2(3), NVARCHAR(120)	No	No overlap (tr_operator_absence_overlap)
execution.service_execution	service_execution_id, booking_service_id, operator_id, execution_status_id	BIGINT, INT, SMALLINT	No	booking_service_id unique
	started_at, finished_at	DATETIME2(3)	Yes	—
	follow_up_notice_at	DATETIME2(3)	Yes	When the notice that the service ended without being recorded was sent; cleared when it is started (changelog 040)
	quality_rating, quality_comment, rated_at	SMALLINT, NVARCHAR(500), DATETIME2(3)	Yes	Rating 1 to 5; comment and date only with a rating

Table	Field	Type	Null	Constraint or description
	is_comment_visible	BIT	No	Audit block
execution.operator_profile	operator_id, specialty	INT, NVARCHAR(80)	No/Yes	PK and FK to operator; operator specialty (changelog 037)
execution.operator_skill	operator_skill_id, operator_id, name, skill_level, display_order	INT, NVARCHAR(60), NVARCHAR(12), SMALLINT	No	FK to operator; unique name per operator; level CERTIFIED, EXPERT, or ADVANCED (ck_operator_skill_level)
execution.inspection_report	inspection_report_id, booking_id, booking_code, vehicle, plate, services, booking_date, booking_time, duration_minutes, public_token, is_published, published_at	BIGINT, NVARCHAR, DATE, TIME, INT, BIT, DATETIME2(3)	No/Yes	One report per booking (uq_inspection_report_booking); unique public token (changelog 033)
execution.inspection_phase	inspection_phase_id, inspection_report_id, phase, status, started_at, finished_at	BIGINT, NVARCHAR(20), DATETIME2(3)	No/Yes	FK to inspection_report; one row per phase (uq_inspection_phase); phase and status checked by ck constraints
execution.inspection_finding	inspection_finding_id, inspection_report_id, phase, area, severity, note	BIGINT, NVARCHAR	No/Yes	FK to inspection_report; severity INFO, MINOR, or MAJOR (ck_inspfinding_severity)
execution.inspection_photo	inspection_photo_id, inspection_finding_id, content_type, content, size_bytes	BIGINT, NVARCHAR(30), VARBINARY(MAX), INT	No	FK to inspection_finding; JPG or PNG up to 2 MB

Note. Source: changelogs 007, 033, 037, and 040 and seed 105 of operations-service.

Table 16

Data Dictionary of the Promotion and Payment Schemas

Table	Field	Type	Null	Constraint or description
promotion.discount_type	discount_type_id, code, name, display_order, is_active	SMALLINT, NVARCHAR	No	PERCENT, FIXED
promotion.promotion	promotion_id, code, name, description, discount_type_id, discount_value, max_discount_amount, min_purchase_amount, is_public, min_completed_booking, valid_from, valid_to, max_redemptions, max_redemptions_per_customer, is_active	INT, NVARCHAR, DECIMAL, BIT, DATE	No/Yes	discount_type PERCENT with discount_value from 1 to 100 (ADR-015)
	price, duration_minutes, icon, featured, benefits	DECIMAL, SMALLINT, NVARCHAR, BIT	Yes	Display fields (changeset 017); price and duration optional
	required_points	INT	No	Points needed to unlock the coupon; 0 = no gate (changeset 018)
promotion.promotion_service, promotion_customer, booking_promotion	Links of a promotion with services, customers, and bookings	INT, BIGINT, DECIMAL	No/Yes	booking_promotion records each redemption (one per coupon and booking)
payment.payment_method_type	payment_method_type_id, code, name, requires_account, requires_receipt, display_order, is_active	SMALLINT, NVARCHAR, BIT	No	EFECTIVO, NEQUI, DAVIPLATA, TRANSFERENCIA
payment.payment_account	payment_account_id, payment_method_type_id	SMALLINT	No	PK; FK to payment_method_type

Table	Field	Type	Null	Constraint or description
	account_holder, account_number, instructions	NVARCHAR(120), NVARCHAR(50), NVARCHAR(300)	No/Yes	—
	qr_image_url	NVARCHAR(MAX)	Yes	QR image as a data URL (changeset 016)
	is_active	BIT	No	Audit block
payment.payment_status	payment_status_id, code, name, is_final, display_order	SMALLINT, NVARCHAR, BIT	No	Ids 1 to 5; 3 = APPROVED
payment.payment	payment_id, booking_id, payment_account_id, payment_status_id	BIGINT, SMALLINT	No	FKs to payment_account and payment_status
	amount	DECIMAL(12,2)	No	> 0; booking total
	processed_at, approved_by, rejection_reason	DATETIME2(3), BIGINT, NVARCHAR(200)	Yes	One approved payment per booking (ux_payment_one_approved_per_booking)
payment.payment_receipt	payment_receipt_id, payment_id	BIGINT	No	FK to payment
	file_url	NVARCHAR(MAX)	No	Receipt image as a data URL (changeset 016)
	transaction_reference, reported_amount	NVARCHAR(100), DECIMAL(12,2)	Yes	Indexed reference (not unique)
	uploaded_at, uploaded_by	DATETIME2(3), BIGINT	No	—
	reviewed_at, reviewed_by, review_comment	DATETIME2(3), BIGINT, NVARCHAR(300)	Yes	Both or neither (ck_preceipt_review)
payment.loyalty_movement_type	loyalty_movement_type_id, code, name, sign, display_order, is_active	SMALLINT, NVARCHAR	No	EARNED, REDEEMED, EXPIRED, ADJUSTED, REVERSED (changeset 020)
payment.loyalty_transaction	loyalty_transaction_id, customer_id, loyalty_movement_type_id, booking_id, points, balance_after, expires_on, description	BIGINT, SMALLINT, INT, DATE, NVARCHAR(200)	No/Yes	FK to payment.loyalty_movement_type; customer_id holds the user_id; EARNED once per customer and booking
	sequence_no	INT	No	Position of the movement in the ledger of the customer; unique per customer (changeset 020)

Note. Source: changelogs 001 to 020 of payment-service. Each table also has the audit block.

Table 17

Data Dictionary of the Notification Schema

Table	Field	Type	Null	Constraint or description
notification.notification_type	notification_type_id, code, name, display_order, is_active	SMALLINT, NVARCHAR	No	26 types (customer, staff, payment, loyalty, inspection, unassigned booking, new app version, service not recorded from changelog 041, and system types)
notification.notification	notification_id, user_id, notification_type_id, title, message, reference_entity, reference_id, is_read, read_at, sent_at	BIGINT, SMALLINT, NVARCHAR(120), NVARCHAR(500), BIT, DATETIME2(3)	No/Yes	FK to notification_type; logical deletion
notification.device_token	Expo push token of each device of a user	—	—	Changeset 016
notification.processed_event	eventId of each consumed event (idempotency)	—	—	Changeset 017
notification.booking_reminder	One pending reminder per booking and lead time (user, booking time, remind_at, sent_at)	—	—	Changeset 019
notification.booking_tracking	booking_id, booking_code, customer_user_id, operator_user_id, scheduled_start, is_cancelled, updated_at	BIGINT, NVARCHAR(20), DATETIME2(3), BIT	No/Yes	Own projection to tell a new booking from a rescheduled one and know the assigned operator (changeset 027)

Note. Source: data model of notification-service. The columns of the last three tables must be completed from their changelogs (Annex B).

11.4 Integrity Rules

Table 18

Integrity Rules and Where They Are Guaranteed

Rule	Where it is guaranteed
Two active bookings cannot occupy the same bay at overlapping times.	Trigger tr_booking_bay_overlap and booking-service.
A vehicle cannot have two overlapping bookings; a booking cannot fall outside the business hours, the break, or the exceptions.	booking-service.
The price of a booking line does not change if the catalog changes.	Trigger tr_service_price_immutable.
A cancellation reason only exists with status CANCELLED.	Constraint ck_booking_cancel.
Only the vehicles of the customer are used in their bookings.	booking-service through customer-service (404 VEHICLE_NOT_FOUND).
The operator assigned must be active, on shift for the whole booking, not absent, and not busy.	AssignmentPolicy in operations-service.
A manual and an automatic assignment at the same time do not give the same operator two services at the same time.	*operations-assignment* lock (*sp_getapplock*) in operations-service.
Only the assigned operator starts or finishes; a service can start from 15 minutes before the booking time and up to 4 hours after its end time (*START_LATE_LIMIT*, since October 24, 2026), so a service that was done can be recorded late (*StartPolicy*, *SERVICE_TOO_EARLY*, *SERVICE_TOO_LATE*; the response carries *startableAt* and *startableUntil*); one service at a time. Starting, finishing, and closing save the execution before calling booking-service and, if booking answers 409 because the change was already applied, treat it as done (OPS-01); executions are locked by row (OPS-03).	operations-service (and booking-service since *develop* of October 23, 2026, E-01).
A booking is rated once, only when COMPLETED, and only by its owner; the rating is written on every line.	operations-service and constraints ck_sexec_*.
At most one approved payment per booking; no second payment while one is in review; the amount is the booking total.	Index ux_payment_one_approved_per_booking and payment-service.
A transaction reference cannot be reused.	payment-service (409 TRANSACTION_REFERENCE_REUSED).
The customer receipt weighs 2 MB at most.	payment-service domain (*Payment.MaxReceiptBytes*, 400 RECEIPT_TOO_LARGE).
A coupon is redeemed only if active, in its dates, and unlocked by the balance; once per booking.	payment-service (CustomerPromotionService) and promotion.booking_promotion.
Two ledger movements of the same customer cannot be computed on the same balance.	Unique sequence_no per customer (409 on conflict).

Note. Prepared by the authors from the code and the data models of the services.

11.5 Migrations and Initial Data

Each service owns the Liquibase changelog of its schemas and applies it at startup; *ddl-auto: validate* then checks that the JPA entities match the tables. Each service records its history in its own tables `DATABASECHANGELOG_<SERVICE>` and `DATABASECHANGELOGLOCK_<SERVICE>`. The payment-service is the exception: its changelogs live in *db/changelog* and are run by the *payment-migrate* container (ADR-013). Every changeset has a precondition *onFail:MARK_RAN*, so a database created with the global script *lavarapido-6-services-sqlserver.sql* converges with one created by the services. The numbering is shared (001 to 013 initial schemas, 014 onward later changes, 101 onward seeds), except in payment-service, which uses its own numbering (TD-019; it is at 023). As of October 26, 2026, the global high-water mark is 049 (booking-service *049-booking-payment-mode-deadline.sql*: *payment_mode*, *payment_deadline*, *payment_reported_at*, and the filtered index *ix_booking_payment_pending*); the next shared

changeset is 050. The latest seed is booking-service 117 (*PAYMENT_TIMEOUT* cancellation reason); the next one is 118. The list of changelogs of each service is in the Installation Manual, section 11.2.

12. Technical Modules

12.1 security-service

Function: identity and access: registration with the acceptance of the legal documents, login with JWT and lockout after failed attempts, sessions, password recovery by code, profile (name, phone, address, email, password, and deactivation), preferences (theme, language, and the three notification switches, with partial update), account administration, and the informative permission matrix of the three fixed roles.

Main files: AuthController (includes *GET /api/v1/auth/session*), ActiveSessionValidator, UserProfileController, UserAdministrationController, InternalUserController; AuthenticationService, AccountRegistrationService, PasswordRecoveryService, EmailChangeService, AccountDeactivationService, UserAdministrationService, UserPreferenceService; PasswordPolicy; JwtTokenIssuer; LoginLockout and LoginLockoutPersistenceAdapter; UserAddressController; EmailResetCodeNotifier; RabbitDomainEventPublisher; DevDataSeeder.

Tables: schemas security and audit.

Rules: passwords with BCrypt (cost 12); password policy of 8 to 72 characters with an uppercase letter, a digit, and a special character, without spaces; the recovery code has 6 digits, lasts 15 minutes, is single use, and five failed attempts invalidate it (counter in memory); requesting a new code invalidates the previous one; the email of a nonexistent account produces the same response; email change and deactivation require the current password; an administrator cannot deactivate their own account (403 CANNOT_DISABLE_OWN_ACCOUNT) and an administrator account cannot be deactivated from user administration (403 ADMIN_CANNOT_BE_DISABLED); the role is set when the account is created and there is no endpoint to change it; five failed attempts within 15 minutes lock the account for 15 minutes (*LoginLockout*, 423 ACCOUNT_LOCKED), and a successful login or a password reset clears the counter; registration stores the acceptance date of the legal documents when *acceptsTerms* and *acceptsDataPolicy* arrive; every token carries its session in the *sid* claim, and since October 21, 2026 *ActiveSessionValidator* rejects the token of a closed session (logout, account disabled by the administrator or deactivated by its owner) on every security-service route; *GET /api/v1/auth/session* answers 204 if the session is still open and 401 if not, for the gateway.

Events: publishes UserRegistered (*security.user_registered*), which carries *createdByAdmin* so that the welcome message explains how to log in; PasswordChanged, EmailChanged, and AccountDeactivated are raised by the domain. **Internal routes:** contact of a user with its switches, and list of active administrators.

12.2 customer-service

Function: customer profile and vehicles.

Main files: CustomerVehicleController, VehicleTypeController, AdminVehicleController; CustomerVehicleService, CustomerRegistrationService (consumes UserRegistered), VehicleTypeQueryService; LicensePlatePolicy and LicensePlate.

Tables: customer.vehicle_type, customer, and customer_vehicle.

Rules: the profile is created from the event UserRegistered or on the first request of the customer (reading *personId* from security-service); plates are stored clean and in uppercase, with the format ABC123 for cars and ABC12D for motorcycles; plates are unique among active vehicles; another customer's vehicle answers 404. **Events:** consumes *security.user_registered* (queue *customer-service.events*); its domain events are only logged.

12.3 booking-service

Function: catalog and prices by vehicle type, establishment, weekly schedule with break, exceptions, bays, availability, and bookings.

Rules: availability is calculated on a 30-minute grid inside the opening hours, outside the break, not in the past, up to 60 days ahead, and with at least one active bay free for the whole duration; a booking is created directly as CONFIRMED with a bay; if the slot is taken, nothing is saved and up to five alternative times of the same day are returned (since October 27, 2026, without those that overlap another active booking of the same vehicle: *AvailabilityPolicy.withoutBusyTimes*, so as not to offer a time that would end in VEHICLE_ALREADY_BOOKED; also when rescheduling); *serviceIds* accepts at most 10 services, without nulls or negatives, and an invalid item of the business hours week answers 400 VALIDATION_ERROR with *errors* per field instead of 500; *bookedBy* (who created the booking) only travels in the staff views (409 SLOT_UNAVAILABLE, which also carries the reason: CLOSED_DAY, OUTSIDE_HOURS, TIME_PASSED, or NO_FREE_BAY, calculated by *AvailabilityPolicy.whyUnavailable*); the request may carry *preferredOperatorId*, which travels in BookingConfirmed so that operations-service respects it if that operator is free; customers can reschedule or cancel only before the start; the administrator moves bookings through IN_PROGRESS, COMPLETED, NO_SHOW, or CANCELLED; each catalog service has its *loyalty_points*, each booking line copies them when the booking is built, and *Booking.totalLoyaltyPoints()* exposes the total in *BookingResponse.totalLoyaltyPoints*; changes of schedule, exceptions, and bays are recorded in *schedule_history*. The business data of the establishment (identity, tax information, and channels) are read and saved with *GET* and *PUT /api/v1/admin/establishment*; the */internal/v1/bookings* routes serve operations-service with the internal key; creating, rescheduling, closing expired bookings, and marking a booking without an operator as NO_SHOW take the schedule lock (*sp_getapplock* on *booking-schedule*). **Expired bookings (since October 21, 2026):** every 5 minutes *ExpiredBookingsJob* calls *BookingService.closeExpired*, which cancels only the bookings still SCHEDULED one hour after their end time (*BOOKING_EXPIRY_GRACE_MINUTES*), with the system reason *EXPIRED* (seed 112), which does not appear in the list of reasons. A CONFIRMED booking is never closed from booking-service, because it may have been served without being recorded: operations-service decides (section 12.4). For one that never had an operator, operations-service calls the internal route *POST /internal/v1/bookings/{id}/no-show* (*BookingService.markUnattended*), which checks again that the booking is CONFIRMED and already expired (409 BOOKING_NOT_EXPIRED otherwise) and moves it to NO_SHOW. Nothing is refunded automatically. The job is turned off with *BOOKING_EXPIRY_CHECK_ENABLED=false* and its interval is *BOOKING_EXPIRY_CHECK_INTERVAL*. **Events:** publishes BookingConfirmed (created or rescheduled) and BookingCancelled (also when an expired booking is cancelled; the move to NO_SHOW publishes no event).

Payment at booking (since October 26, 2026): the customer must send *paymentMode* (*DIGITAL* or *CASH*) when creating the booking (400 *INVALID_PAYMENT_MODE*); the administrator does not have to, and their bookings have no deadline or limit. *PaymentPolicy* applies two rules: at most *BOOKING_CASH_ACTIVE_LIMIT* (1) active (SCHEDULED, CONFIRMED, or IN_PROGRESS) cash bookings made by the customer themselves (*booked_by*), counted with the schedule locked (*lockSchedule*; 409 *CASH_BOOKING_LIMIT*), and, for *DIGITAL*, *paymentDeadline* = min(creation + *BOOKING_PAYMENT_WINDOW* (30 min), start). *PaymentEventsListener* consumes *payment.reported* and *payment.confirmed* and sets *payment_reported_at* (*Booking.markPaymentReported*, idempotent). *PaymentDeadlineJob* runs every *BOOKING_PAYMENT_EXPIRY_INTERVAL* (60 s; turned off with *BOOKING_PAYMENT_EXPIRY_ENABLED=false*) and, before cancelling an overdue booking (*Booking.paymentOverdue*), asks payment-service: with *reported=true* it marks it and does not cancel it; with *false* it cancels it with

PAYMENT_TIMEOUT and publishes *BookingCancelled*; if payment does not answer, it leaves it for the next round (*PaymentTimeoutOutcome*: CANCELLED, PAYMENT_REPORTED, or NOT_OVERDUE). Rescheduling does not change the mode or the deadline, and earlier bookings have *payment_mode* NULL and none of these rules.

12.4 operations-service

Function: operators, weekly shifts, absences, operator profile and skills, manual and automatic assignment, execution, ratings, and inspection reports (ADR-014).

Rules: the operator rows are created from the accounts with the OPERATOR role; if two requests create the same operator at the same time, the second one reads the row already created instead of failing; *AssignmentPolicy* returns the reason that blocks an operator (OPERATOR_INACTIVE, OPERATOR_NOT_ON_SHIFT, OPERATOR_ABSENT, OPERATOR_BUSY); only SCHEDULED or CONFIRMED bookings are assigned, and one *service_execution* is created per booking line. **Automatic assignment (RF-008):** when BookingConfirmed is consumed, *AutoAssignmentPolicy* chooses the operator requested by the customer if still free and, otherwise, the free operator with the fewest bookings that day (on a tie, the lowest id); if nobody is free, the booking is left without an operator for the administrator, and a reschedule keeps the operator it already had. *GET /api/v1/operators/available* lists the free operators the customer can choose when booking. A service can be started from 15 minutes before the booked time (*StartPolicy*); starting and finishing move the booking in booking-service; the assignments returned by *GET /api/v1/admin/assignments* include the bay of the booking (ADR-015). Since October 21, 2026 manual and automatic assignment take the same lock (*ExecutionRepository.lockAssignments*, *sp_getapplock* on *operations-assignment*) before reading the operator bookings and saving, and *GET /api/v1/admin/assignments/unassigned* returns *{count, bookingIds}*, the single count of bookings without an operator (SCHEDULED or CONFIRMED, with no operator on any line, and whose time has not ended) used by the dashboard and the bookings screen of the web and the mobile app. **Bookings that ended without being closed (since October 21, 2026):** every 5 minutes *FollowUpJob* calls *FollowUpService.followUp*, which reviews the bookings of the last 7 days with *FollowUpPolicy*. The policy only decides on its own when it is sure: NO_OPERATOR (CONFIRMED without an operator, one hour after its end) is closed as NO_SHOW through the internal route of booking-service; NOT_STARTED (CONFIRMED with an operator, not started when its time ended) and NOT_FINISHED (IN_PROGRESS not finished one hour after its end) are never marked on their own and stay "to close". For these two, operations-service publishes ServiceNotStarted or ServiceNotFinished only once and stores the time of the notice in *execution.service_execution.follow_up_notice_at* (changelog 040), which is cleared when the service is started. *GET /api/v1/admin/assignments/to-close* lists the bookings to close (by default, the last 28 days, *FOLLOW_UP_LOOK_BACK_DAYS*) and *POST /api/v1/admin/assignments/{bookingId}/attended* (*FollowUpService.closeAsAttended*) closes one as attended: the open lines become COMPLETED without made-up hours (without a start time it does not count toward punctuality and without an end time it adds no worked hours, but it does count as a service done and as income), the booking moves to IN_PROGRESS if needed and then to COMPLETED, and ServiceCompleted is not published, so the customer does not get "ready for pickup" hours later; the customer can still rate it. If the booking is not to close, the answer is 409 BOOKING_NOT_TO_CLOSE. The administrator's "No-show" is still the status change of booking-service (*PATCH /api/v1/admin/bookings/{id}/status*) and only applies to a booking that was never started. The operator's lists no longer show NO_SHOW or CANCELLED bookings. The job uses *FOLLOW_UP_ENABLED*, *FOLLOW_UP_INTERVAL*, and *BOOKING_EXPIRY_GRACE_MINUTES*, and needs *INTERNAL_API_KEY*. **Inspection (RF-036):** only for services of 180 minutes or more; phases Reception, Exterior, Interior, Engine, and Delivery; up to five photos of 2 MB per finding, stored in *execution.inspection_photo*; the public link uses a random token and only works after publishing. **Events:** consumes

booking.confirmed and publishes OperatorAssigned, ServiceStarted, ServiceCompleted, InspectionShared, ServiceNotStarted, and ServiceNotFinished.

12.5 payment-service

Function: payment accounts with QR, payments with receipt, review, refunds, and manual payments (ADR-008 and ADR-013).

Rules: only bookings CONFIRMED, IN_PROGRESS, or COMPLETED can be paid, and *PaymentPolicy.CanReportPayment* also requires that there is no approved payment and none pending or in review (exposed as *payable* in *GET /api/v1/payments/me/bookings*); the amount due is the total returned by booking-service minus the redeemed coupon (*AmountDueCalculator*); the payment amount is calculated by the server (*AmountDueCalculator*, in *PaymentCommandService* when reporting and when recording an in-person payment), not by the screen; what the customer says they paid is stored separately in the receipt (*reported_amount*) so that the administrator can compare it, and the payment goes directly to IN_REVIEW; a reused transaction reference is rejected; the receipt weighs 2 MB at most (400 RECEIPT_TOO_LARGE; the web and the app compress it before sending it); *GET /api/v1/payment-accounts* returns the active accounts, including the cash account while it is active; rejection requires a reason of up to 200 characters; a manual payment is approved immediately; the state machine is PENDING → IN_REVIEW → APPROVED or REJECTED, and APPROVED → REFUNDED. **Loyalty and promotions (ADR-015):** an approval credits the points of the booking as EARNED (idempotent per customer and booking); a refund reverses them (REVERSED); the balance is the *balance_after* of the last movement; a coupon is a percentage discount unlocked when the balance reaches *required_points*, and redeeming it does not consume points. **Events:** publishes PaymentConfirmed, PaymentRejected, PaymentRefunded, PromotionRedeemed, and LoyaltyPointsEarned after the commit, only with *MESSAGING_ENABLED=true*. **Correlation and health:** *CorrelationIdMiddleware* takes or creates the *X-Correlation-Id* and forwards it to booking-service; */health* and */actuator/health* check SQL Server with a 3-second limit.

Payment at booking (since October 26, 2026): when the customer reports a payment with a receipt (*POST /api/v1/payments*, *Payment.ReportWithReceipt*), also when reporting it again after a rejection, payment-service publishes *PaymentReported* (*payment.reported*; payload *paymentId*, *bookingId*, *customerUserId*, *amount*). The internal path *GET /internal/v1/payments/bookings/{bookingId}/reported* answers *{"bookingId":123,"reported":true}*, with *reported* true if there is an IN_REVIEW or APPROVED payment; it is protected by *InternalApiKeyAuthenticationHandler* (*X-Internal-Key = INTERNAL_API_KEY*; 401 without a key or with a wrong one, and a user's JWT does not work). Until that day payment-service did not validate any internal key. Reporting or approving the payment of a cancelled booking still answers 409 *BOOKING_NOT_PAYABLE* (*BookingPaymentGuard.EnsureBookingIsPayable*).

12.6 notification-service

Function: converts events into notifications (inbox, push through Expo, and email) and schedules booking reminders (ADR-011).

Rules: the notification is always saved first, and a push failure never makes the request fail; consumption is idempotent by *eventId* (*processed_event*); BookingConfirmed schedules reminders 24 h and 1 h before, BookingCancelled cancels them, and a rescheduling moves them; customer emails read the current address and switches from security-service through the internal route; a notice is emailed only when it changes something for the customer (booking, reminder, payment result, coupon, and inspection report), while service progress, points earned, and every staff notice stay in the inbox and push; the projection *booking_tracking* feeds the staff notices (rescheduled, cancelled, and unassigned for the operator; created, rescheduled, and cancelled for every active administrator); the administrator endpoint accepts only *SYSTEM_MESSAGE* and *INSPECTION_REPORT*; InspectionShared creates the notice to the

customer with the public link of the report (*/report/{token}*); every few minutes *UnassignedBookingAlertService* warns the administrators about confirmed bookings that start in less than 2 hours and still have no operator (once per booking); the internal route *POST /internal/v1/notifications/app-update*, called by *deploy/publish-apk.sh*, announces by push that a new version of the application is available; since October 21, 2026, ServiceNotStarted and ServiceNotFinished (*execution.service_not_started* and *execution.service_not_finished*) create *OPERATOR_SERVICE_NOT_REGISTERED* for the operator ("¿Atendiste este servicio?", did you serve this booking?) and *ADMIN_SERVICE_NOT_REGISTERED* for every active administrator ("Servicio sin registrar", service not recorded), only in the inbox and by push, with the types seeded in changelog 041, and a *null* field in an event payload is ignored instead of sending the whole message to the DLQ. **Configuration:** MESSAGING_ENABLED, PUSH_ENABLED, MAIL_ENABLED, INTERNAL_API_KEY, REMINDERS_ENABLED, REMINDER_LEADS, and REMINDER_CHECK_INTERVAL.

12.7 Web and Mobile Clients

The web application consumes the gateway through the services of *core/services*; *auth-interceptor* adds the token and redirects to */auth?expired* when the server answers 401 while the session is still open (a 401 that arrives after logging out is ignored, so "Your session expired" is not shown to someone who left on their own), and *api-error* translates the error codes. Since October 21, 2026 the *auth* service listens to the *storage* event: if another tab of the same browser signs out, this one goes back to sign-in, and if another account signs in, it reloads so it does not show the previous account's data; the payment screen compresses the receipt to at most 2 MB and enlarges the QR with a magnifier. Every screen reads and saves through the API: the business data use *BusinessStore* over booking-service, and the inspection reports use *inspection-api* over operations-service (the photos are compressed in the browser before being sent). Only the reports are calculated in the browser from the bookings and payments; since October 21, 2026 only completed bookings count as services done. Roles and permissions, promotions, loyalty, preferences, and notification switches are read and saved through the API (*role-permissions-api*, *promotions-api*, *loyalty-api*, and *preferences*). The mobile application covers the three roles with equivalent screens and stores the session with *expo-secure-store*. Its HTTP client (*core/api/httpClient.ts*) sends JSON or *multipart* forms (the inspection photos), accepts its own timeout per request (60 seconds to upload photos), and applies the same rule as the web to a 401. *core/api/config.ts* exposes *WEB_URL*, the address of the web used to build the public link of the report; it is taken from *EXPO_PUBLIC_WEB_URL* or, when it is not set, from the same host as the gateway with port 4200. Since the night of October 21, 2026 the administrator's home, on the web (*reservations-store*) and in the app, shows the "To close" card with the "Attended" and "No-show" buttons only when there are bookings to close, and both use *GET /api/v1/admin/assignments/unassigned* for the count of bookings without an operator (if operations-service does not answer, they apply the same rule with the data they have). In the app, also, the administrator's booking form saves the chosen operator and status, the customer's points bar is full when nothing is left to unlock and does not show "0% OFF", the operator sees "—" as the rating when there are no ratings and the operator's home no longer repeats the "In progress" counter, the roles tab says that the three roles are fixed and the permissions are for reference, and the new notices open the operator's services screen or the administrator's home.

12.8 Life Cycles

Table 19

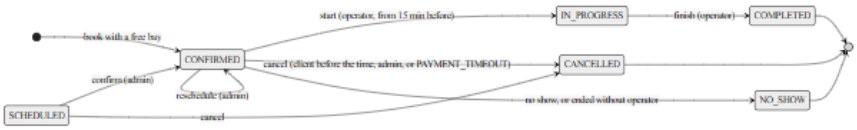
Domain State Machines

Entity	States (database codes)	Main transitions
Booking	SCHEDULED, CONFIRMED, IN_PROGRESS, COMPLETED, CANCELLED, NO_SHOW	Creation → CONFIRMED; SCHEDULED → CONFIRMED; CONFIRMED → IN_PROGRESS → COMPLETED; SCHEDULED or CONFIRMED → CANCELLED or NO_SHOW. The system only closes a booking when it is sure: one hour after its end, CONFIRMED without an operator → NO_SHOW, and a CONFIRMED booking with digital payment that reaches its deadline without a receipt → CANCELLED with the PAYMENT_TIMEOUT reason. The automatic cancellation of an expired SCHEDULED booking (EXPIRED) was removed on October 24, 2026 (B-07). A CONFIRMED booking with an operator that nobody recorded does not change on its own: the administrator resolves it (Attended → IN_PROGRESS → COMPLETED, or No-show).
Service execution	PENDING, IN_PROGRESS, PAUSED, COMPLETED, WITH_ISSUE	PENDING → IN_PROGRESS → COMPLETED. PAUSED and WITH_ISSUE exist in the catalog but are not used.
Payment	PENDING, IN_REVIEW, APPROVED, REJECTED, REFUNDED	PENDING → IN_REVIEW → APPROVED or REJECTED; APPROVED → REFUNDED.
Point movement	EARNED, REVERSED (also REDEEMED, EXPIRED, and ADJUSTED in the catalog)	Payment approval → EARNED; refund → REVERSED. Movements are never modified.

Note. Prepared by the authors from the code and the data models of the services.

Figure 3

Booking State Diagram



Note. Gray states are final. The administrator can also change the status of a booking directly from the panel (*PATCH /api/v1/admin/bookings/{id}/status*). *PAYMENT_TIMEOUT* is the cancellation done by *PaymentDeadlineJob* (section 12.3) and *ended without operator* is requested by *FollowUpJob* of operations-service (section 12.4); *ExpiredBookingsJob*, which cancelled expired scheduled bookings, no longer exists (B-07). A confirmed booking with an operator that nobody recorded does not change status on its own: the administrator resolves it from "To close" (Attended → IN_PROGRESS → COMPLETED, or No-show).

13. API or System Routes

The external API follows the REST style (Fielding, 2000). All external access goes through the gateway. The OpenAPI contracts of the seven components are in *07-api/contracts/openapi* of the documentation, and each Java service publishes its Swagger UI in the *dev* profile (*/swagger-ui.html*); payment-service publishes it at */swagger* in the Development environment.

13.1 General Conventions

- **Versioning:** prefix */api/v1* in the path.
- **Format:** JSON; numeric identifiers (IDENTITY), not UUIDs.
- **Authentication:** header *Authorization: Bearer `<JWT>`* on every non-public route.
- **Correlation:** header *X-Correlation-Id*.
- **Ownership:** the identity comes from the verified token, never from the body; prices, totals, and end times are never accepted from the client.
- **Pagination:** the paginated lists (accounts and notifications) return a *PageResponse* object with the items and the paging data.
- **Errors:** RFC 9457 problem details with a stable *code* (section 9.5.3).
- **Internal routes:** */internal/v1/** with the header *X-Internal-Key*; they are not routed by the gateway.

13.2 Authentication in the API

security-service issues a JWT signed with HS256 (secret JWT_SECRET of at least 32 bytes) valid for one hour, with the claims *sub* (user identifier), *roles*, *sid* (session identifier), *iss* (lavarapido-security-service), *aud* (lavarapido-api), *iat*, *nbf*, *exp*, and *jti*. There is no refresh token: when the token expires, the person logs in again. A JWT cannot be revoked by itself before its expiration (Jones et al., 2015). Since October 21, 2026 the system makes up for it with the *sid* claim: logging out, or the administrator disabling the account or the owner deactivating it, closes the session record; security-service rejects the tokens of that session and the gateway asks it (*GET /api/v1/auth/session*) before passing any request on to the other services, keeping the answer 30 seconds per token (section 9.2). Every service validates the signature, the issuer, and the audience with the same secret.

13.3 Endpoints

Table 20

Endpoints of the System

Method	Route	Role	Description
POST	/api/v1/auth/register	Public	Registers a customer (documentNumber, firstName, lastName, email, phone, password, acceptsTerms, acceptsDataPolicy).
POST	/api/v1/auth/login	Public	Returns accessToken, expiry, and user; 423 ACCOUNT_LOCKED after five failed attempts.
POST	/api/v1/auth/logout	Token	Closes the session; the token stops working.
GET	/api/v1/auth/session	Token	204 if the session of the token is still open; 401 if it was closed. Used by the gateway.
POST	/api/v1/auth/password/forgot, /verify, /reset	Public	Password recovery in three steps.
GET, PATCH	/api/v1/users/me	Token	Profile.
PUT	/api/v1/users/me/email, /api/v1/users/me/password	Token	Change email or password (current password required).
POST	/api/v1/users/me/deactivate	Token	Deactivates the own account.
GET, PUT	/api/v1/users/me/address	Token	Contact address of the profile.
GET, PUT	/api/v1/users/me/preferences	Token	Theme, language, and the switches notificationsEnabled (push), emailRemindersEnabled, and promotionsEnabled; PUT is partial.
GET, POST	/api/v1/admin/users	ADMIN	Lists (paginated, by role) and creates accounts.
PATCH	/api/v1/admin/users/{userId}/status	ADMIN	Activates or deactivates an account.
GET	/api/v1/admin/roles-permissions	ADMIN	Permission catalog and the permissions of each fixed role.

Method	Route	Role	Description
PUT	/api/v1/admin/roles-permissions/{role}	ADMIN	Replaces the permissions of a role (400 UNKNOWN_PERMISSION).
GET	/api/v1/vehicle-types	Public	Vehicle types.
GET, POST	/api/v1/vehicles	CLIENT	Own vehicles; registers a vehicle.
GET, PUT, DELETE	/api/v1/vehicles/{id}	CLIENT	Own vehicle (404 if it belongs to another customer).
GET	/api/v1/admin/vehicles?ids= or ?plate=	ADMIN	Vehicles with their ownerUserId.
GET	/api/v1/catalog/categories, /api/v1/catalog/services?vehicleTypeId=	Public	Catalog with prices by vehicle type.
GET	/api/v1/schedule/business-hours, /api/v1/schedule/exceptions, /api/v1/establishment	Public	Schedule and data of the business.
GET	/api/v1/bookings/availability?date=&vehicleTypeId=&serviceIds=	Token	Free start times of a day.
POST	/api/v1/bookings	CLIENT	Creates a booking.
GET	/api/v1/bookings/me, /api/v1/bookings/{id}, /api/v1/bookings/cancellation-reasons	CLIENT	Own bookings and catalog of reasons (without the EXPIRED and PAYMENT_TIMEOUT system reasons).
PUT, POST	/api/v1/bookings/{id}, /api/v1/bookings/{id}/cancel	CLIENT	Reschedule or cancel before the start. The web and the app only use the cancellation (without a reason: CUSTOMER_REQUEST is stored); rescheduling is done by the administrator.
GET, POST	/api/v1/admin/bookings (?from=&to=&status=)	ADMIN	Lists and creates bookings for any customer.
GET	/api/v1/admin/bookings/day-summary?date=	ADMIN	Indicators of the day for the dashboard.
GET, PUT	/api/v1/admin/bookings/{id}	ADMIN	Detail and rescheduling.
PATCH	/api/v1/admin/bookings/{id}/status	ADMIN	Changes the status (with reasonCode when canceling).
GET, POST, PUT, DELETE, PATCH	/api/v1/admin/catalog/services[/{id}[/status]]	ADMIN	Services and prices.
PUT	/api/v1/admin/schedule/business-hours	ADMIN	Weekly schedule.
GET	/api/v1/admin/schedule/history	ADMIN	History of changes to the schedule, exceptions, and bays.
GET, PUT	/api/v1/admin/establishment	ADMIN	Business data of the establishment.
POST, PUT, DELETE	/api/v1/admin/schedule/exceptions[/{id}]	ADMIN	Exceptions.
GET, POST, PUT, DELETE	/api/v1/admin/bays[/{id}]	ADMIN	Bays.
GET, PATCH	/api/v1/operator/bookings[/{id}[/status]]	OPERATOR, ADMIN	Used by operations-service.
GET	/api/v1/admin/operators	ADMIN	Operators.
GET, PUT	/api/v1/admin/operators/profiles, /api/v1/admin/operators/{id}/profile	ADMIN	Specialty and skills of the operators.
GET	/api/v1/operators/available?date=&time=&minutes=	Token	Free operators the customer can choose when booking.
PATCH, PUT	/api/v1/admin/operators/{id}/status, /availability	ADMIN	Activate or deactivate; replace the weekly shift.
GET, POST, DELETE	/api/v1/admin/operators/{id}/absences[/{absenceId}]	ADMIN	Absences.
GET	/api/v1/admin/assignments (?from=&to=)	ADMIN	Assignments.

Method	Route	Role	Description
GET	/api/v1/admin/assignments/unassigned (?from=&to=)	ADMIN	*{count, bookingIds}* of the scheduled or confirmed bookings without an operator that have not ended yet; the dashboard and the bookings screen of the web and the app use this count.
GET	/api/v1/admin/assignments/to-close (?from=&to=)	ADMIN	Bookings with an operator that ended without being recorded (NOT_STARTED or NOT_FINISHED); by default, the last 28 days (*FOLLOW_UP_LOOK_BACK_DAYS*).
POST	/api/v1/admin/assignments/{bookingId}/attended	ADMIN	Closes a booking to close as attended, without made-up hours (409 BOOKING_NOT_TO_CLOSE if it is not).
PUT	/api/v1/admin/assignments/{bookingId}	ADMIN	Assigns or changes the operator (409 with the reason).
GET	/api/v1/admin/assignments/{bookingId}/candidates	ADMIN	Operators with their blocking reason.
GET, POST, PATCH, DELETE	/api/v1/admin/inspections/{bookingId}[/phases/{phase}, /findings[/{findingId}], /publish, /share, /photos/{photoId}]	ADMIN	Inspection report: start, phases, findings with photos (multipart), publish, and notify the customer.
GET	/api/v1/public/inspections/{token}[/photos/{photoId}]	Public	Published report and its photos, without logging in.
GET	/api/v1/operator/services (?from=&to=)	OPERATOR	Own services.
POST	/api/v1/operator/services/{bookingId}/start, /finish	OPERATOR	Starts or finishes a service.
GET	/api/v1/operator/ratings	OPERATOR	Ratings received.
GET, POST	/api/v1/ratings/me, /api/v1/ratings/{bookingId}	CLIENT	Ratings given; rates a booking (1 to 5, comment up to 500).
GET	/api/v1/payment-accounts	Token	Active accounts with QR; the cash account only appears while it is active.
POST, GET	/api/v1/payments, /api/v1/payments/me	CLIENT	Reports a payment with a receipt of at most 2 MB (400 RECEIPT_TOO_LARGE); own payments.
GET	/api/v1/payments/me/bookings	CLIENT	Latest payment status and *payable* flag of each own booking.
GET, POST	/api/v1/admin/payments (?status=)	ADMIN	Lists payments; registers a manual payment.
GET	/api/v1/admin/payments/amount-due	ADMIN	Real amount due of a booking (with the coupon discount).
POST	/api/v1/admin/payments/{id}/approve, /reject, /refund	ADMIN	Review of a payment.
GET, POST, PUT	/api/v1/admin/payment-accounts[/{id}], /api/v1/admin/payment-methods	ADMIN	Payment accounts and methods.
GET, POST, PUT, PATCH, DELETE	/api/v1/admin/promotions[/{id}[/active]], /api/v1/admin/promotions/metrics	ADMIN	Promotions (coupons), pause and resume, and metrics.
GET	/api/v1/loyalty/balance, /api/v1/loyalty/promotions	CLIENT	Points balance and promotions with their *unlocked* flag.
POST	/api/v1/loyalty/redeem	CLIENT	Redeems a coupon on a booking.
GET	/api/v1/notifications, /api/v1/notifications/unread-count	Token	Inbox (filters and pagination) and counter.
PATCH, POST, DELETE	/api/v1/notifications/{id}/read, /unread, /read-all, /{id}	Token	Read status and deletion.
PUT, POST	/api/v1/notifications/devices, /devices/unregister	Token	Expo push tokens.
POST	/api/v1/admin/notifications	ADMIN	Manual notification; only SYSTEM_MESSAGE or INSPECTION_REPORT (400 NOTIFICATION_TYPE_NOT_ALLOWED). No screen of the web or the app uses it.
GET	/internal/v1/users/{userId}/contact	Internal key	Contact and notification switches of a user for notification-service.

Method	Route	Role	Description
GET	/internal/v1/users/admins	Internal key	Identifiers of the active administrators.
GET	/internal/v1/bookings?from=&to=, /internal/v1/bookings/{id}	Internal key	Bookings for the automatic assignment of operations-service.
POST	/internal/v1/bookings/{id}/no-show	Internal key	Moves to NO_SHOW a confirmed booking that ended without an operator; only operations-service calls it (409 BOOKING_NOT_EXPIRED if it has not expired yet).
POST	/internal/v1/notifications/app-update	Internal key	Push notice of a new version of the mobile application.
GET	/internal/v1/payments/bookings/{bookingId}/reported	Internal key	Since October 26, 2026, payment-service: whether the booking has an IN_REVIEW or APPROVED payment (*{"bookingId":123,"reported":true}*); only booking-service calls it before cancelling for lack of payment.
GET	/actuator/health	Public	Health of each service and of the gateway.

Note. Prepared by the authors from the controllers of the services and the service documentation.

13.3.1 Creating a Booking

The body of the request uses local dates of the business. The server calculates the end time, the bay, and the price.

```
POST /api/v1/bookings
Authorization: Bearer <JWT>
Content-Type: application/json

{
  "vehicleId": 12,
  "serviceIds": [1],
  "date": "2026-10-12",
  "time": "10:30",
  "notes": "Check the upholstery",
  "paymentMode": "DIGITAL"
}
```

Note. Structure of *CreateBookingRequest* of the web application; the values are fictitious.

Table 21

Relevant Error Responses When Creating a Booking

Situation	Response
The vehicle does not exist or belongs to another customer	404 VEHICLE_NOT_FOUND (the same response on purpose).
customer-service does not answer	503 CUSTOMER_SERVICE_UNAVAILABLE.
The time cannot be booked	409 SLOT_UNAVAILABLE with up to five alternative times the vehicle can take and the reason: CLOSED_DAY, OUTSIDE_HOURS, TIME_PASSED, or NO_FREE_BAY.
The vehicle already has a booking at that time	409 VEHICLE_ALREADY_BOOKED.
The service is not offered for the vehicle type	SERVICE_NOT_AVAILABLE_FOR_VEHICLE.
Invalid date (past or beyond 60 days)	INVALID_DATE.
Invalid shape of the request	400 VALIDATION_ERROR.
The customer did not send *paymentMode* or it is neither *DIGITAL* nor *CASH* (since October 26, 2026)	400 INVALID_PAYMENT_MODE.
The customer already has *BOOKING_CASH_ACTIVE_LIMIT* active cash bookings	409 CASH_BOOKING_LIMIT.

Note. Codes taken from the translation file of the web application and the documentation of booking-service.

13.4 Domain Events (RabbitMQ)

The messaging uses a single topic exchange, *carwash.events*. The routing key follows the convention `<owner-schema>.<event_in_snake_case>` (Hohpe & Woolf, 2003). Every message carries the same envelope:

```
{
  "eventId": "uuid (idempotency key)",
  "eventType": "BookingConfirmed",
  "aggregateId": "145",
  "occurredAt": "2026-10-12T15:00:00Z",
  "version": 1,
  "payload": { "bookingId": 145, "customerUserId": 7, "scheduledStart": "2026-10-12T15:00:00Z" }
}
```

Table 22

Events Published and Consumed

Event	Routing key	Publisher	Consumers	Status
UserRegistered	security.user_registered	security-service	customer-service, notification-service	Published
BookingConfirmed	booking.confirmed	booking-service	notification-service, operations-service (automatic assignment)	Published
BookingCancelled	booking.cancelled	booking-service	notification-service	Published; it carries *reason* and, since October 24, 2026, *reasonCode*
BookingStatusChanged	booking.status_changed	booking-service	notification-service	Published when the administrator changes the status (since October 23, 2026); with IN_PROGRESS, COMPLETED, or NO_SHOW it cancels the reminders, and NO_SHOW has its notice (*BOOKING_NO_SHOW*, since October 24)
OperatorAssigned	execution.operator_assigned	operations-service	notification-service	Published
OperatorUnassigned	execution.operator_unassigned	operations-service	notification-service	Published since October 24, 2026 when a rescheduled booking has no free operator: it notifies the released operator and the administrators
ServiceStarted	execution.service_started	operations-service	notification-service	Published
ServiceCompleted	execution.service_completed	operations-service	notification-service	Published
InspectionShared	execution.inspection_shared	operations-service	notification-service	Published (with the token of the public link)
ServiceNotStarted, ServiceNotFinished	execution.service_not_started, execution.service_not_finished	operations-service	notification-service	Published once per booking (since October 21, 2026)
PaymentReported	payment.reported	payment-service	booking-service	Published since October 26, 2026, when a payment with a receipt is reported
PaymentConfirmed	payment.confirmed	payment-service	notification-service; booking-service since October 26, 2026	Published
PaymentRejected, PaymentRefunded	payment.rejected, payment.refunded	payment-service	notification-service	Published
PromotionRedeemed	payment.promotion_redeemed	payment-service	notification-service	Published

Event	Routing key	Publisher	Consumers	Status
LoyaltyPointsEarned	payment.loyalty_points_earned	payment-service	notification-service	Published (with unlockedPromotions)
BookingCreated	booking.created	—	—	Not published; since October 24, 2026 notification-service no longer listens to it (H-12)

Note. Delivery is at least once; acknowledgment after processing; three retries and then the dead-letter queue (*.dlq*); publication after the commit, without Outbox. Since October 24, 2026 operations-service consumes from *operations-service.booking-confirmed* with its *.dlq* (the old *operations-service.events* queue is drained and deleted on startup) and publishes with broker confirmation; notification-service stores in the *pending_delivery* table, in the same transaction, the email and push it sends after the commit, and retries them (1, 2, 4… minutes, at most 1 hour, up to 5 attempts). Each event that generates a notification carries the user identifier of the recipient (*customerUserId* or *operatorUserId*), so notification-service does not query other services (ADR-011). Since October 26, 2026 booking-service consumes from *booking.payment-events* (keys *payment.reported* and *payment.confirmed*), with its dead-letter queue *booking.payment-events.dlq* through the *carwash.events.dlx* exchange. Since there is no Outbox, a *payment.reported* can be lost; that is why *PaymentDeadlineJob* confirms with payment-service before cancelling (AC-89).

13.5 Health and Verification

Each service and the gateway expose */actuator/health* (and payment-service also */health*). All of them check the database: payment-service waits at most 3 seconds for SQL Server and, if it does not answer, replies DOWN with code 503. On the public server, *docker-compose.prod.yml* also checks the health of every container every 30 seconds (*docker compose ps* shows them *healthy* or *unhealthy*). The recommended integration checks are described in the Installation Manual, section 16, and the end-to-end tests in *lavarapido-infra/e2e* run the complete flows against the containers; they run only locally, because the CI only checks their syntax.

14. Security

The security policy of the project is based on defense in depth, least privilege, secure defaults, security by design, and zero trust, and its checklist follows the OWASP Top 10 (OWASP Foundation, 2021).

14.1 Authentication and Session Control

Table 23

Authentication Parameters

Element	Decision
Token	Bearer JWT issued by security-service.
Algorithm	HS256 with a shared secret (minimum 32 bytes); RS256 is the planned evolution.
Validity	One hour (JWT_ACCESS_TOKEN_TTL); no refresh token.
Claims	sub, roles, sid, iss, aud, iat, nbf, exp, and jti; no name, email, phone, or document.
Passwords	BCrypt with cost 12 (Provos & Mazières, 1999); maximum 72 characters, the input limit of BCrypt.
Login attempts	Five failed attempts within 15 minutes lock the account for 15 minutes (423 ACCOUNT_LOCKED, table *security.login_lockout*); a successful login or a password reset clears the counter; the response time is the same for unknown emails.
Sessions	security.user_session records each login; logout sets revoked_at, and deactivating or disabling the account closes all its sessions.
Revocation	Since October 21, 2026 the token of a closed session stops working: security-service rejects it by its *sid* claim (*ActiveSessionValidator*) and the gateway cuts it before the other services (*SessionCheckFilter*, answer kept 30 seconds per token; if security-service does not answer, the request goes on: accepted risk D-02). Since October 24, 2026 the filter checks every request with *Authorization*, with any scheme (*bearer*, *Bearer*, *BEARER*), and signing out clears the cache in all its variants (SEC-01).

Note. Source: configuration and code of security-service and ADR-006.

14.2 Authorization

The system has three fixed roles (ADMIN, OPERATOR, and CLIENT) and no custom roles; the tables *permission* and *role_permission* hold an informative permission matrix that the administrator edits, but no route checks it yet (ADR-015). Each service declares in its security configuration which roles reach each route (for example, */api/v1/admin/** requires ADMIN). A valid role is not enough: each service also verifies **ownership**, so an operator cannot start the service of another and a customer reaches only their own bookings, vehicles, and payments. Calls between services forward the token of the user, so the target service authorizes the original user. The routes */internal/** use a different filter (*InternalApiKeyFilter*) with the key *INTERNAL_API_KEY*; if the key is empty, the routes stay closed.

14.3 Input Validation

- Validation at the edge with Bean Validation (Java) and in the domain with value objects (EmailAddress, DocumentNumber, PhoneNumber, LicensePlate, PersonName); the validations of the forms are only a convenience.
- Allowlists instead of blocklists (for example, @gmail.com emails, Colombian plate formats, and four themes).
- **SQL injection:** JPA and EF Core with parameters; concatenating SQL is forbidden.
- **XSS:** Angular interpolation escapes by default; the sanitizer is not bypassed.
- Prices, totals, and end times never come from the client.

14.4 Secrets and Communication

- The secrets (sa password, JWT secret, SMTP, RabbitMQ, and internal key) are read from environment variables or from *lavarapido-infra/.env*, excluded from Git; the repository only contains *.env.example*.
- In the local environment the services connect with *sa*; before any shared deployment, each service must have a login with rights only on its schemas.
- HTTPS with TLS 1.2 or later is mandatory outside the local environment; only the gateway must be exposed. On the public server, since October 20, 2026 Caddy adds the HSTS, *X-Content-Type-Options: nosniff*, *X-Frame-Options*, *Referrer-Policy*, and *Permissions-Policy* headers, removes the *Server* header, and sends with the web a content security policy (CSP) that only allows the project's own code, Google Fonts, and jsDelivr.
- The demonstration accounts of *DevDataSeeder* exist only in the *dev* profile.

14.5 OWASP Top 10 Checklist

Table 24

Controls by OWASP Category

Category	Control	Status
A01 Broken access control	Role per route and ownership in each service; identity from the token.	Applied
A02 Cryptographic failures	BCrypt; secrets in environment variables; TLS outside local.	Applied (TLS with Let's Encrypt on the public server)
A03 Injection	Parameterized queries; validation of inputs.	Applied
A04 Insecure design	Same response for unknown emails; identifiers are not used as access control.	Applied
A05 Security misconfiguration	No stack traces in responses; Swagger only in dev; security headers and CSP on the public server.	Applied
A06 Vulnerable components	Review with versions:display-dependency-updates, dotnet list package --vulnerable, and npm audit.	Manual
A07 Identification and authentication failures	One-hour token; the token of a closed session stops working with any scheme; temporary lockout after failed attempts, also for the recovery code and the current password; a limit of 60 requests per minute per IP on */api/v1/auth/** with the client's real IP (since October 24, 2026); consent required at sign-up.	Applied
A08 Software and data integrity failures	Pinned versions; validation of messages; invalid messages to the DLQ.	Applied
A09 Security logging and monitoring failures	Logs without personal data and with correlation id; audit trail not populated.	Partial
A10 Server-side request forgery	The services only call configured internal addresses.	Applied

Note. Prepared by the authors from the security policy and the code.

14.6 Personal Data Protection

The processing of personal data must comply with Statutory Law 1581 of 2012 (Congreso de la República de Colombia, 2012). Registration requires the explicit acceptance of the Terms and Conditions and of the Personal Data Processing Policy (two checkboxes on the web and on the mobile app). *RegisterRequest* sends *acceptsTerms* and *acceptsDataPolicy*, and security-service stores the date and time of each acceptance in *security.app_user* (columns *terms_accepted_at* and *data_policy_accepted_at*, changelog 039). No card data are collected. The only personal data traveling on the message bus are the email and the name in UserRegistered, and its *toString()* omits them so that they do not reach the logs.

14.7 Accepted Risks

- The token of a closed session still works for up to 30 seconds (the answer the gateway keeps) or, if security-service does not answer, until it expires.
- Symmetric HS256 signature with a secret shared among services.
- The request limit only covers */api/v1/auth/**, lives in the gateway's memory (per replica), and does not group IPv6 by /64; the rest of the API has no limit.
- Possible loss of events between the commit and the publication (no Outbox, TD-011).
- Shared SQL Server instance and *sa* account in the local environment.
- Role permissions editable but not enforced (ADR-015).
- Display fields and discount fields of a promotion not kept consistent by any constraint (ADR-015).

15. Version Control

15.1 Source Code

The project uses Git with repositories on GitHub. Work is done on *feature/**, *feat/**, or *fix/** branches that are merged into *develop* (integration); *main* holds stable versions. On October 23, 2026 *develop* was merged into *main* and tagged *v1.0.0* in 8 repositories (web, api-gateway, security, customer, booking, operations, notification, and infra); payment-service got it on October 24 (PR #2 merged), and that same day *develop* moved ahead again with the audit fixes. Those fixes shipped on October 26, 2026 in *v1.1.0*, in the 10 repositories, including the mobile app, whose first version this is (PR #1 "Release v1.1.0"). That same day *v1.1.1* shipped in the web and the mobile app (terms and conditions in the third person, with the section on the inspection report and its photographs), in payment-service (README test counts), and in infra (payment-service is deployed from *main*, like the others). On October 27, 2026 *v1.1.1* shipped in customer-service and booking-service (the validations and alternatives of October 27, *bookedBy* only for the staff, and the single owners query) and *v1.1.2* in the web (WebP backgrounds), payment-service (integration tests in the CI), infra (the payment method in the end-to-end tests), and the mobile app (a corrected text, published over the internet). That same day payment-service *v1.1.3* was merged into *main* (required credentials and no passwords in the code); its tag and release were published on November 4, 2026. *main* is protected in the 10 repositories: it only takes changes through a pull request approved by another member with a green CI. Commits follow the format *type(scope): description* (for example, *feat(payment): JWT auth, payment accounts with QR, admin review and manual payments*). The conventions are in *00-governance/git-conventions.md*.

15.2 Database Schema

The schema is versioned with Liquibase: each change is a changeset with an identifier and an author, and Liquibase stores a checksum. Changesets are immutable and forward-only; a later change is recorded in a new changeset.

15.3 Architecture Decisions

Table 25

Architecture Decision Records

ADR	Topic	Status
ADR-001	Documentation language (English)	Accepted
ADR-002	Architectural style and original decomposition	Superseded by ADR-009
ADR-003	Database strategy: one instance, one owner per schema	Accepted
ADR-004	Communication between services: REST and RabbitMQ	Accepted
ADR-005	API Gateway	Accepted
ADR-006	Authentication and authorization: HS256 JWT verified by each service	Accepted
ADR-007	Internal architecture: hexagonal	Accepted
ADR-008	Payment verification by QR and receipt	Accepted
ADR-009	Revised decomposition into six services	Proposed (in use)
ADR-010	Alignment of the data model and the frontend	Accepted
ADR-011	Notifications: push, events, emails, and reminders	Accepted
ADR-012	Booking: vehicle resolution over REST	Accepted
ADR-013	payment-service in .NET 8 with layered architecture	Proposed (in use)
ADR-014	Operations: booking integration, operator provisioning, and manual assignment	Proposed (in use)
ADR-015	Loyalty points, promotions, fixed-role permissions, and per-booking bays	Accepted

Note. Source: folder *05-architecture/decisions/records* of the documentation.

15.4 Continuous Integration and Deployment

Continuous integration. Each of the 10 code repositories has *.github/workflows/ci.yml*, which runs on every push or pull request to *develop* and *main*: *./mvnw verify* in the Java services and the gateway, *dotnet build* and *dotnet test* in payment-service, the production build and *ng test* in the web, *tsc --noEmit* and *npm test -- --ci* in the mobile app, and *shellcheck*, *actionlint*, *docker compose config*, and the syntax of *e2e/* and *e2e/remote/* in *lavarapido-infra*. As of October 24, 2026 the latest *develop* run is green in all 10. The *main* branch is protected (required review and CI) in Brayan Cabrera's repositories, customer, and booking; payment-service and the mobile app have no protection, and *develop* is not protected in any of them.

Deployment. The Azure server is updated with *deploy/update-server.sh* (backup, *git pull* of each repository, *docker compose up -d --build*). Since October 24, 2026, before rebuilding it keeps the running version as *:prev*, then waits up to 8 minutes for the 11 containers to be healthy, restarts Caddy if its *Caddyfile* changed, and, if everything is healthy, cleans unused images; *deploy/rollback.sh* goes back to *:prev* without building, and *setup-server.sh* never downgrades (*merge-base --is-ancestor*; *ALLOW_DOWNGRADE=1* allows it on purpose). That day *develop* was deployed in two runs over SSH (the second with *BRANCH=develop*, which was needed until *v1.1.0*): 11 healthy containers and migrations 044, 045, 113, 047, 116, 022, and 023 applied (Deployment Document, section 1.3.9). *main* also has the "Desplegar en Azure" workflow (*deploy.yml*), which logs in over SSH with a key limited to *deploy/ci-deploy.sh*; since October 26, 2026 it has its secrets and runs on its own on every merge into *main* (continuous deployment, below).

Continuous deployment (active since October 26, 2026). *lavarapido-infra* (*006b5b0*) and the "Pedir despliegue" job of the 8 repositories deployed to Azure (web, api-gateway, security, customer, booking, operations, notification, and payment) make every approved merge into *main* deploy on its own: the repository's CI, if it passes, launches *deploy.yml* with *automatic=true* (secret *DEPLOY_DISPATCH_TOKEN*); the workflow backs up, pulls *main* of every repository (until *v1.1.0*, payment-service stayed on *develop* through *DEVELOP_ONLY*; since infra *v1.1.1* it also pulls *main*), waits up to 20 minutes for another update in progress, requires all containers to become healthy within 8 minutes, restarts Caddy if its *Caddyfile* changed, and checks the web and the API. With the machine off it leaves a notice and does not fail; when it is turned on, *deploy/boot-update.sh* (an *@reboot* task) deploys what is pending by comparing with *~/lavarapido/backups/deployed-heads*. The mobile app has *cd.yml*: on each push to *main* it repeats the CI, compares the code's native fingerprint with that of the published APK and, if it matches, publishes the over-the-air update (*eas update*, secret *EXPO_TOKEN*); if it changed, it asks for a new APK. To activate it, *AZURE_SSH_KEY*, *DEPLOY_DISPATCH_TOKEN* (in customer, booking, and payment their owner or an administrator creates them), and *EXPO_TOKEN* were created, and *v1.1.0* was released on October 26, 2026, with *lavarapido-infra* last (Deployment Document, section 4.3.4). It was closed as AC-90 in the corrective actions document. **First automatic run (October 26, 2026, night, Colombia time).** With the secrets created, merging the "Release v1.1.1" PR into *main* of *lavarapido-infra* launched the **Desplegar en Azure** workflow with *automatic=true* at 22:19, and the following requests from the other repositories deployed on their own until 22:48 (those that arrived while another one was running were cancelled and served by the next one): all of them ended green, with "Listo: todos los servicios están sanos" and "Web y API responden 200". Since the 22:46 run, payment-service also pulls *main*. In the mobile app, publishing *v1.1.0* failed because *EXPO_TOKEN* was missing (nothing was published); with the secret created, APK *versionCode* 6 was published at 22:39 and the *v1.1.1* run compared the native fingerprint of the commit with that APK's, saw that it matched, and published the over-the-air update (Deployment Document, section 1.3.12). **Runs of October 27, 2026.** The merges of the web *v1.1.2* (01:03, Colombia time) and of customer and booking *v1.1.1* and payment and infra *v1.1.2* (01:35 and 01:36) deployed on their own and ended green, with «Web y API responden 200»; at 14:30, the mobile app *v1.1.2* was published over the internet after checking that the native fingerprint matched that of APK *versionCode* 6

(Deployment Document, section 1.3.13). The deployment requested by the merge of payment-service *v1.1.3* (17:29) found the machine off and deployed nothing; that release installed itself when the machine was turned on, on November 4, 2026 (*deploy/boot-update.sh*, Deployment Document, section 1.3.14). **SonarCloud (October 26, 2026).** The findings of the analysis were fixed: in api-gateway, a real null access in the authentication path and comparing CIDR networks by content; in security, operations, and notification, the RabbitMQ publishers and the SMTP and Expo senders marked as *@Primary* (rule *java:S9352*); in the web, a stricter CI, the Angular CLI pinned, and sorting with *localeCompare*; and in the mobile app, promises marked with *void*, callbacks wrapped in arrow functions, the shared *Pill* component, the password translation key renamed, and the CI scripts and workflows hardened.

16. Maintenance Recommendations

16.1 Local Environment

The local environment and its startup order are described in the Installation Manual. SQL Server and RabbitMQ must be healthy before the services; each service must tolerate that a dependency is not ready and answer 503 with a stable code.

16.2 Configuration

The configuration variables are described in the Installation Manual, section 10. Any new variable must be added to *.env.example* with a comment and, if the service runs in Docker, to its *environment* block of *docker-compose.yml*. Those of October 21, 2026 (*SESSION_CHECK_** in the gateway, *BOOKING_EXPIRY_** in booking-service, with *BOOKING_EXPIRY_GRACE_MINUTES* shared with operations-service, and *FOLLOW_UP_ENABLED*, *FOLLOW_UP_INTERVAL*, and *FOLLOW_UP_LOOK_BACK_DAYS* in operations-service; nine in total) are already in those files (action ACO-14, closed). Those of October 23, 2026 (*ALLOWED_EMAIL_DOMAINS* and *DEMO_*_PASSWORD* in security-service) are too. *ADMIN_DOCUMENT* and *ADMIN_PHONE* complete the data of the first administrator in production. Those of October 26, 2026 (booking-service's *BOOKING_CASH_ACTIVE_LIMIT*, *BOOKING_PAYMENT_WINDOW*, *BOOKING_PAYMENT_EXPIRY_ENABLED*, *BOOKING_PAYMENT_EXPIRY_INTERVAL*, and *PAYMENT_SERVICE_URL*, and *INTERNAL_API_KEY*, which payment-service now also needs and *docker-compose.prod.yml* requires) are in *.env.example* and in the compose file.

16.3 Updates and Dependencies

- Before each delivery, run *./mvnw versions:display-dependency-updates*, *dotnet list package --outdated --vulnerable*, and *npm outdated* and *npm audit*, and fix critical and high vulnerabilities.
- Pin exact versions; do not use floating versions.
- Keep the ADRs, the data models, and the OpenAPI contracts updated when the design changes; export the contracts from the running service (*/v3/api-docs*).

16.4 Database

- A new domain table must create its *tr_touch_* trigger in the same changeset.
- Every new query must include *deleted_at IS NULL*.
- Seeds use fixed identifiers; never reseed without *IDENTITY_INSERT* (CANCELLED = 5, APPROVED = 3).
- The next changeset of the global numbering is 050, and the next seed is 118 (as of October 26, 2026); payment-service is at its 023.
- Back up the database before deploying a service that adds changesets.

16.5 Testing

The testing strategy proposes a pyramid of unit, integration, contract, and end-to-end tests, with at least 80% line coverage and 90% in the domain. As of October 27, 2026, the *main* CI of each repository passed 1,348 tests: web 242 in 68 files, gateway 62, security-service 200, customer-service 83, booking-service 137, operations-service 110, notification-service 133, payment-service 275 with *v1.1.3* (142 domain, 61 application, 47 API, and 25 integration tests, which the "Pruebas de integración" job runs against SQL Server 2022 with the Liquibase migrations, without skipping any; on a machine without a database the 12 that use SQL Server are skipped), and the mobile app 106 with *jest-expo*; the 19 end-to-end tests passed 19 of 19 locally that day. As of October 24, 2026, the latest *develop* CI of each repository had passed 1,119 tests: web 212 in 66 files, gateway 62, security-service 200, customer-service 82, booking-service 95, operations-service 110, notification-service 133, and payment-service 225 (134 domain, 52 application, 34 API, and 5 infrastructure; the 11 persistence tests used to be skipped in the CI because they need the isolated *LavaRapidoPaymentIT* database; since October 27, 2026 payment's CI has the "Pruebas de integración" job, which starts SQL Server 2022, applies the Liquibase migrations like *payment-migrate*, and runs them); the mobile app has 73 tests in 25 suites with *jest-expo*: 1,192 in total. *lavarapido-infra/e2e/remote* adds a battery of 44 cases against the public server (*e2e.yml*, by hand), tested only against a simulator. On October 23, 2026 there were 838. On October 27, 2026 booking-service has 137 and customer-service 83 (the administrator's vehicle list reads the owners in a single query), and the end-to-end tests send the payment method when booking. payment-service release *v1.1.3* was merged into *main* on October 27, 2026 (PR #16), and its tag and release were published on November 4; it has been in production since November 4 (15.4): no passwords in the code, the service does not start without *DB_PASSWORD* and, with messaging on against a remote RabbitMQ, not without *RABBITMQ_USERNAME* and *RABBITMQ_PASSWORD* either (*guest* only with a local RabbitMQ); its local test database moved to SQL Server 2022 and *LavaRapidoPaymentIT*, like the CI, and it has 275 in the CI (domain 142, application 61, API 47, and 25 in the infrastructure project, 12 of them against SQL Server). The previous complete run, on the night of October 21, 2026, gave these results: 707 unit and integration tests passed, none skipped (web 62 in 41 files, gateway 40, security-service 155, customer-service 69, booking-service 50, operations-service 67, notification-service 114, and payment-service 150: 115 domain, 28 application, and 7 persistence tests against the isolated *LavaRapidoPaymentIT* database), 18 more than in the earlier run of the same day because of the new classes *FollowUpPolicyTest* and *FollowUpServiceTest* (operations) and *DomainEventEnvelopeTest* (notification), the internal no-show route in *BookingServiceTest* and *BookingApiWebTest*, and "To close" in *reservations-store.spec.ts*; also, 19 of 19 end-to-end tests, with the gateway session check on. Until October 23, 2026 the mobile app had no automated tests and its continuous integration only checked the types (*npx tsc --noEmit*); the 42 functional cases were last run on October 19, 2026 and all passed against the complete system (*lavarapido-infra/e2e*: complete booking, payment, and service flow, inspection, and access rules). The Java services test their domain, use cases, and web layer without a database (*./mvnw test*); payment-service has four test projects (domain, application, API, and integration). There are still no persistence tests against a real SQL Server inside the Java services (TD-010); the end-to-end tests partly cover that gap. The recommended concurrency tests are two simultaneous bookings for the same bay, two updates with the same *row_version*, two simultaneous point movements of the same customer, and the duplicate delivery of the same event.

16.6 Diagnosing Frequent Problems

Table 26

Frequent Problems and Suggested Checks

Symptom	Probable cause	Check
401 with a token that seemed valid	Expired token, different JWT_SECRET, issuer, or audience, or closed session (logout on another device or disabled account).	Log in again; verify the .env file; *GET /api/v1/auth/session* with the token tells whether the session is still open.
400 RECEIPT_TOO_LARGE when reporting a payment	The receipt weighs more than 2 MB (a client without the compression of the web or of the updated app).	Upload it from the web or from the updated app, which compress the image.
A booking moved by itself to No-show or Cancelled	One hour passed after its end and it was confirmed without an operator (*FollowUpJob*), or it had digital payment and reached its deadline without a receipt (*PaymentDeadlineJob*). *ExpiredBookingsJob* no longer exists (B-07).	Status and reason of the booking (PAYMENT_TIMEOUT on those cancelled for lack of payment); booking-service log (*Cancelled N booking(s) for payment timeout*) and operations-service log (*Follow-up: N booking(s) closed as not attended*).
A booking appears in "To close"	It had an operator and nobody started it, or it was started and nobody finished it.	*GET /api/v1/admin/assignments/to-close*; status of its *service_execution* rows and *follow_up_notice_at*.
503 CUSTOMER_SERVICE_UNAVAILABLE or BOOKING_SERVICE_UNAVAILABLE	The dependency is down or its URL is wrong.	Health of the service and CUSTOMER_SERVICE_URL or BOOKING_SERVICE_URL.
"Customer profile not ready" (CUSTOMER_NOT_PROVISIONED)	The UserRegistered event was not consumed.	Queue customer-service.events and its DLQ.
Notifications that do not arrive	MESSAGING_ENABLED=false, an event without recipient, or a message in the DLQ.	Configuration and queue notification-service.events.dlq.
Compose build fails for operations-service	The repository was not cloned as lavarapido-operation-service.	Installation Manual, section 8.
Points not credited after an approval	Services with 0 points, MESSAGING_ENABLED=false (no notice), or BOOKING_SERVICE_URL wrong.	payment.loyalty_transaction and the log of payment-service.
Push not delivered	PUSH_ENABLED=false, Expo Go, or a deactivated token.	Configuration and notification.device_token.
Liquibase checksum error	An executed changeset was edited.	Revert and create a new changeset.

Note. Prepared by the authors from the runbooks of the services.

16.7 Technical Debt

Table 27

Main Technical Debt Items

Item	Description	Priority
TD-007	Forward X-Correlation-Id from customer-service and notification-service to security-service and in the messages; generate it in the gateway.	Low
TD-008	Optimistic locking with @Version.	Low
TD-010	Persistence tests against a real SQL Server (Testcontainers).	Low
TD-011	Outbox for events.	Low
TD-012	Metrics with Micrometer and Prometheus.	Low
TD-019	payment changelogs with their own numbering.	Low

Note. Source: *15-project-control/technical-backlog.md* of the documentation (Vehicle Washing Development Team, 2026a), reviewed against the code on October 16, 2026. Between October 8 and 16, 2026, TD-001, TD-002, TD-003, TD-009, TD-013, TD-014, TD-015, TD-016, TD-017, TD-018, TD-020, and TD-021 were resolved; their closure is recorded in the corrective, preventive, and improvement actions document. TD-006 was also closed on October 19, 2026, with continuous integration in GitHub Actions. On October 21, 2026, TD-024 (tokens of closed sessions), TD-025 (bookings nobody closed and the count of bookings without an operator), and TD-026 (an operator assigned twice) were closed; on the night of the same day the rule of TD-025 was adjusted so that a booking with an operator does not end as No-show unless the administrator decides it; and TD-027 (tests of the administrator screens) was resolved for bookings, management, and reports.

16.8 Recommended Evolution

If the project leaves the academic scope, the documentation identifies as natural evolution: RS256 signature, refresh tokens, a request rate limit across the whole API (today only on */api/v1/auth/**), Outbox, one database login per service (or one database per service), audit triggers, real enforcement of the permission matrix, mTLS between services, observability (metrics and tracing), image storage outside the database, and a payment gateway. Each change requires an ADR.

17. Glossary

Table 28

Technical Glossary

Term	Definition
ADR	Architecture Decision Record.
Aggregate	Group of domain objects kept consistent together, with a root (Evans, 2004); for example, Payment.
API Gateway	Single external entry point that routes the requests; in this system it does not check the token signature, but since October 21, 2026 it cuts the tokens of closed sessions.
Hexagonal architecture	Ports and adapters style in which the domain does not depend on the infrastructure (Cockburn, 2005).
Bay	Physical washing station; the number of active bays is the capacity.
Correlation id	Identifier that follows a request through the gateway, the services, and the messages.
Dead-letter queue (DLQ)	Queue where the messages that failed after the retries are kept.
Domain event	Business fact published so that other services react.
Idempotency	Processing the same event more than once without duplicate effects.
JWT	JSON Web Token: signed token that carries identity and roles (Jones et al., 2015).
Ledger	Record of the point movements of each customer; the balance is that of the last movement.
Liquibase	Tool to version and migrate the database schema.
Logical deletion	Mark *deleted_at* instead of deleting the row.
Outbox	Pattern that stores events in the same transaction before publishing them; not adopted.
Problem details	Standard format of HTTP errors (RFC 9457) with a stable code.
Topic exchange	RabbitMQ exchange that routes messages by routing keys.
Value object	Object defined by its values (for example, LicensePlate or EmailAddress).

Note. Prepared by the authors from the glossary of the project and the references cited.

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18. Annexes

Annex A. Source of the Figures

Figures 1 and 3 are diagrams prepared from the code. Figure 2 was generated on October 16, 2026, from the real database, with all the services started, so it shows the 57 tables that existed that day; later *notification.pending_delivery* (migration 047) was added, and today there are 58. No figure contains secrets or personal data.

Annex B. Discrepancies Detected

Table 29 gathers the contradictions found between the code, the project documentation, and version 2.0 of the manuals. Each one must be resolved before the final delivery.

Table 29

Discrepancies and Decision Adopted in This Manual

Topic	Discrepancy	Criterion used here
Stack of the services	Version 2.0 stated Java 21 and Spring Boot 3 in the six services.	Code: Spring Boot 4.0.8 in the Java services and .NET 8 in payment-service (ADR-013).
Migrations	Version 2.0 described a single master changelog for the whole database.	Code and migration strategy: one changelog per service applied at startup; payment through payment-migrate.
Gateway	Version 2.0 said that the gateway rejects invalid JWTs at the edge.	Gateway documentation and ADR-006: it does not check the token signature; each service does. Since 2026-10-21 it does cut the tokens of closed sessions, asking security-service.
Account lockout	Version 3.1 stated that it did not exist.	Code of 2026-10-16: five failed attempts within 15 minutes lock the account for 15 minutes (RF-004b). Resolved.
Operator assignment	Version 3.1 stated manual assignment only and no choice by the customer.	Code of 2026-10-16: automatic assignment when the booking is confirmed, with the optional choice of the customer (RF-008 and RF-010). Resolved.
Close of bookings nobody recorded	Version 4.5 said that every CONFIRMED booking nobody started became NO_SHOW on its own one hour after its end.	Code of the night of 2026-10-21: only a CONFIRMED booking without an operator changes on its own; one with an operator stays "to close" for the administrator (*FollowUpPolicy*). Resolved.
Token revocation	Up to version 4.4 a token kept working until its expiration after logout.	Code of 2026-10-21: the token of a closed session stops working in security-service and in the gateway (TD-024). Resolved.
Audit trail	The architecture says that triggers feed audit.audit_log.	Code: the table exists but no trigger or service writes to it.
Booking request	Version 2.0 used customerVehicleId, scheduledStart, and services with quantity.	Code: vehicleId, serviceIds, date, time, and notes.
Execution codes	Version 2.0 used Spanish codes (PENDIENTE, EN_PROCESO...).	Data model: PENDING, IN_PROGRESS, PAUSED, COMPLETED, WITH_ISSUE.
Number of tables	Version 2.0 cited 43, 44, or 49 tables; version 3.1 cited 49.	57 tables on 2026-10-16, checked in the database (the tables for account lockout, person address, operator profile and skills, and the four inspection tables were added); 58 since notification migration 047 (*pending_delivery*), counted in the real database on 2026-10-27.
Error format	Version 2.0 described an object with error, message, and traceId.	Code: RFC 9457 problem details with code.
Promotions	Version 3.0 stated that promotions worked only in the browser.	Code of 2026-10-09: percentage coupons in payment-service, unlocked with points (ADR-015). Resolved.
Payment notices and points	Version 3.0 stated that payment-service did not publish events.	Code of 2026-10-09: it publishes five events and credits points (ADR-011 and ADR-015). Resolved.
Repeated reference	ADR-008 and RF-016 forbid reusing a transaction reference.	Code of 2026-10-09: rejected with TRANSACTION_REFERENCE_REUSED. Resolved.
Name of the operations repository	Version 3.0 of the manuals called it lavarapido-operations-service.	Real name: lavarapido-operation-service (expected by docker-compose.yml). Resolved.
Administration module	Version 2.0 of the administrator manual stated that it had no code.	Code: complete module under /admin.
README of lavarapido-api-gateway	It said that the gateway did not answer CORS and did not list the routes of operations-service and payment-service.	Fixed on 2026-10-21: it answers CORS (*CorsConfig*), cuts the tokens of closed sessions, and lists every route. Resolved.
README of lavarapido-infra	It marked operations-service and payment-service as planned.	Fixed on 2026-10-16: every service appears as implemented. Resolved.
Team guide (section 6)	It listed as mock data screens that were already connected.	Fixed on 2026-10-16: only the reports, calculated in the browser, and the permission matrix, which is not enforced, remain. Resolved.

Topic	Discrepancy	Criterion used here
Interface texts	The landing page announced a service "wherever you are", the "Report progress" button sent nothing, the "Delete account" option only deactivated the account, and the refund texts were only in Spanish.	Fixed on 2026-10-16: the page talks about the service at the car wash, the button was removed, the option is called "Deactivate my account", and the refund is translated into the four languages. Resolved.
Dictionary of the notification schema	The columns of device_token, processed_event, and booking_reminder were not available in the reviewed copy.	Purpose documented; columns to be completed.

Note. Prepared by the authors from the cross-analysis of the code and the documentation.

Annex C. Delivery Checklist

Table 30

Technical Manual Checklist

Criterion	Met
Complete title page and version	Yes / No
Table of contents with updated pages	Yes / No
Objective and scope explained	Yes / No
The content matches the code of the final version	Yes / No
Entity-relationship diagram included	Yes / No
Project structure verified against the real code	Yes / No
Discrepancies of Annex B resolved	Yes / No
No credentials, keys, or sensitive data	Yes / No
Spelling and APA formatting reviewed	Yes / No
Exported to PDF with the name Manual_tecnico_LavadoVehicular_3145556.pdf	Yes / No

Note. Adapted from the checklist of the guide for writing software manuals (SENA, 2026).